A review of Biomarkers of Environmental Enteric Dysfunction in children under 5 years in Low- and Middle-Income Countries

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DECLARATION

Declaration by Candidate

This thesis is my original work and has not been presented for a degree in any other University or any other award.

Name	**Signature**	**Date**
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Declaration by Supervisor(s)

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DEDICATION

I dedicate this work to my parents who have been supportive of me in all my visions.

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I would like to express my heartfelt gratitude to the Almighty Creator, Pwani University, The KEMRI-Wellcome Trust Research Programme, and The Initiative to Develop African Research Leaders (IDeAL) Training Department for granting me this invaluable opportunity to learn and conduct research at such a prestigious institution. I am also deeply thankful to my parents for their emotional support, to my supervisors for their guidance and patience during the refinement of my research skills, and to my friends and colleagues Lina Akinyi, Brenda Kamau, Elisha Omer, and Justin Getonto for their camaraderie and collaboration.

ABSTRACT

Background: Environmental enteric dysfunction (EED) is a subclinical chronic disease of the small intestines characterized by gut and systemic inflammation, villus blunting, malabsorption and gut permeability. EED is widespread among children and adults in low- and middle-income countries (LMICs) driven by enteric pathogens, contaminated environments, and microbiota dysbiosis. Childhood EED is associated with stunting, unresponsiveness to oral vaccines, slowed cognitive development, and micronutrient deficiencies. Diagnosis of EED remains challenging, partly due to lack of standardized non-invasive biomarkers in LMICs.

Objectives: This study aimed to review the distribution of widely used biomarkers of EED including Calprotectin (CAL), Myeloperoxidase (MPO), Alpha-1-antitrypsin (AAT), Neopterin (NEO), and the dual sugar gut permeability test, in studies conducted in LMICs among children under five years of age.

Methods: A comprehensive search was conducted on PubMed, Google Scholar, and Web of Science databases. A PRISMA-based approach using Covidence, an online software, was adopted for screening of the studies. Studies where the selected biomarkers were examined among children under 5 years in LMIC settings were included. Forest plots were used to visualize the distribution of these biomarkers across various geographical sites, countries, settings including rural and urban, and child health status and identify any cutoffs used.

Results: Of the 402 publications identified through database searches, 35 studies from 22 countries and involving a total of 17,430 children were included. The sample sizes varied across studies and sites, ranging from 19 to 1,512 participants. There was also considerable variability in age and health status of the children included. No study determined an optimal cutoff for these biomarkers in LMIC settings. Studies employed

biomarkers cutoffs that would trigger investigations for inflammatory bowel disease in high income settings to indicate inflammation (MPO<2000ng/mL, CAL<50µg/g, NEO<70nmol/L) and permeability (AAT<270µg/g, and L:M<0.07). The median concentrations for MPO, NEO, CAL, AAT, and L:M were above these cutoffs in 81%, 100%, 71%, 84%, and 83% respectively of the included studies.

Conclusions: These studies highlight the need to establish cutoffs relevant for children in these settings. Biomarker cutoffs adopted would trigger investigations for inflammatory bowel disease in high-income countries and therefore study show high levels of EED biomarkers among children under five in LMICs. Interventions targeting EED could have population-wide impacts on growth and development in LMICs.

ABBREVIATIONS

1. AAT- Alpha-1-antitrypsin
2. APCs- Antigen Presenting Cells
3. CAL- Calprotectin
4. CI- Confidence Interval
5. EED- Environmental enteric dysfunction
6. IQR- Interquartile range
7. L:M- Lactulose to Mannitol Ratio
8. LMICs- Low- and Middle-Income Countries
9. MPO- Myeloperoxidase
10. NEO- Neopterin
11. SD- Standard Deviation

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CHAPTER 1: INTRODUCTION

1.0 Background

The 2022 UNICEF-WHO-World Bank Joint Malnutrition Estimates revealed that 148 million children worldwide, under five years old, are stunted and 45 million are wasted. Approximately 52% and 43% of children under 5 years old in Asia and Africa respectively, are stunted (*Levels and Trends in Child Malnutrition*, 2023). It is estimated that 6% annual average reduction rate is required to curb stunting to 88.9 million by 2030 (*Levels and Trends in Child Malnutrition Child*, 2023). Bhutta and colleagues indicate that even if the existing nutrition-specific interventions were implemented globally with approximately 90% coverage, stunting prevalence would only be reduced by 20% (Bhutta et al., 2013). The expected poor performance is partly due to the complex interactions between infections and undernutrition in low- and middle-income countries (LMIC) settings (M. Kosek et al., 2014).

Children living in low-resource settings are repeatedly exposed to multiple enteric pathogens from poor sanitary conditions and faecally contaminated environments including food and water (Kelly et al., 2004; Kotloff et al., 2013). Such exposures often results in gut damage currently known as environmental enteric dysfunction (EED) (McCormick BJJ et al., 2019; Watanabe & Petri, 2016). Environmental enteric dysfunction was previously known as tropical enteropathy (Fagundes-Neto et al., 1984), tropical sprue (Keusch et al., 2013; Mathan, 1990), or subclinical enteropathy (Haghighi et al., 1997). It has also been referred to as malnutrition enteropathy (Sturgeon et al., 2023).

EED is an asymptomatic disorder of the small intestine characterized by marked mucosal and systemic inflammation, reduced intestinal absorptive capacity, reduced intestinal

barrier integrity, and bacterial translocation (Arndt et al., 2016). Histologic findings in EED include chronic inflammation, villus blunting, crypt hyperplasia, and reduced villus to crypt ratio (Mahfuz et al., 2024; Mathan, 1990). Recent comparative histological analysis of intestinal biopsies from Bangladesh, Pakistan, Zambia, and the United States has revealed that EED has other key features in addition to reduced surface area and inflammation. These include a markedly reduced number of goblet cells and Paneth cells, both of which make a major contribution to barrier function by maintaining a protective mucus on the epithelial surface, which has antimicrobial properties (Gizaw et al., 2022; Liu et al., 2020). Additionally, crypt branching and intramucosal infiltration of Brunner's glands also characterize EED (Hodges et al., 2025)

The subclinical nature of EED was, in the past, misconstrued and thought to be harmless, since there were no obvious clinically relevant symptoms seen among those histologically diagnosed (Watanabe & Petri, 2016). Currently, EED has been implicated in many conditions, including stunting, wasting, poor neurocognitive development, micronutrient deficiencies, low vaccine efficacy, enteric microbiome dysbiosis, and increased risk of child mortality in LMICs (R. Goto et al., 2009; Naylor et al., 2015). For example, 43% of growth faltering among Gambian children was attributed to EED (Lunn et al., 1991). Additionally, a recent prospective cohort study linked biomarkers of EED to worse motor development among rural Bangladeshi children (Ho et al., 2025).

Though research has clearly indicated EED is a key factor in the aetiology of poor childhood outcomes, its diagnosis remains challenging (Gilmartin & Petri, 2015a). Upper gastrointestinal endoscopy with biopsy is considered the gold standard in EED diagnosis (Fagundes-Neto et al., 1984; Keusch et al., 2013). However, it requires specialised skills and is not ethically feasible in well children under the age of 5 years due to its invasive

nature (Watanabe & Petri, 2016). Moreover, endoscopy is barely available in most LMIC settings and is unlikely to be used in large-population studies (Tickell et al., 2019). This has therefore necessitated the need for the identification of other biomarkers that are less invasive but also widely applicable in low-resource settings where the burden is greatest.

There are a variety of biomarkers targeting EED pathways that have been evaluated as diagnostics for EED (Tickell et al., 2019). These biomarkers are less invasive than endoscopy, and are drawn from a variety of body compartments, including urine, stool, and blood (Crane et al., 2015; van der Merwe LF et al., 2013). However, to fully realize their utility, it is essential to establish widely accepted diagnostic criteria that incorporate these biomarkers (Tickell et al., 2019; Watanabe & Petri, 2016).

The dual sugar intestinal permeability test is a non-invasive technique that uses two sugar molecules to assess permeability and absorption aspects of the small bowel mucosa (Menzies et al., 1979). One sugar reflects total small bowel absorptive capacity, and is most often mannitol, or occasionally rhamnose. Lactulose, or sometimes lactose, is often used to indicate small bowel permeability (Weser & Sleisenger, 1965). The widely used combination is the lactulose to mannitol (L:M) test (Cobden et al., 1978). Xylose has also been used to assess intestinal permeability in the LMICs, although its validity is critiqued due to possible hepatic metabolism and jejunal absorption (Denno et al., 2016). However, few laboratories in LMICs have the necessary expertise and equipment to perform this assay. Additionally, the dual sugar test is reported to be more expensive than other biomarker tests measured from stool and blood (Kosek et al., 2013).

Other biomarkers used include Myeloperoxidase (MPO), Calprotectin (CAL), Neopterin (NEO), and Alpha-1 Antitrypsin (AAT), among others (Campbell et al., 2004; M. Kosek et al., 2013; Prendergast et al., 2015a). Myeloperoxidase serves as a specific marker for

neutrophil activity, indicating inflammation within the intestinal mucosa (McCormick et al., 2017). Calprotectin, a marker of neutrophil activity in the intestinal mucosa, is a calcium- and zinc-binding protein predominantly found in neutrophils (Tøn et al., 2000). Neopterin is produced by macrophages or dendritic cells in response to interferon-gamma (IFN-g) stimulation during pro-inflammatory responses mediated by Th1 lymphocytes (Campbell et al., 2004). Alpha-1-antitrypsin is a large molecular weight protein mainly synthesised in the liver, and serves as a marker for intestinal permeability (Strygler et al., 1990). However, these biomarkers have not been adequately standardised for diagnostic purposes, which poses a significant challenge to fully realising their potential in this context (Wen et al., 2023). Further, there is controversy on the optimal cutoffs for these biomarkers, especially among children under five (Wen et al., 2023).

Many studies have relied on cutoffs reported from research and clinical settings involving populations in high-income countries (HICs) (Campbell RK et al., 2017; Gizaw et al., 2022; M. Kosek et al., 2013; Lin A et al., 2020). The commonly used cutoffs for these biomarkers are <0.27mg/g for AAT, < 2000 ng/ml for MPO, < 70 nmol/l for NEO and <0.07 for L:M ratio (Jukic et al., 2021; K. Kolho et al., 2012; M. Kosek et al., 2013; Vaos et al., 2013). While these cutoffs have not been systemically derived, the studies have indicated that sick and well children in LMICs often have higher average concentrations of these biomarkers than the cutoffs among affluent populations in HICs.

1.1 Statement of the Problem

There are currently no clear diagnostic criteria for EED, which poses a significant challenge in understanding its role in stunting and assessing treatment and prevention strategies (Harper et al., 2018). Although small bowel histological examination is regarded as the gold standard for diagnosis, its use is limited in population studies. It is

highly invasive, expert-dependent, and ethically difficult in asymptomatic individuals, making it impractical for routine diagnosis (Prendergast et al., 2015b). Additionally, the availability of endoscopy-guided biopsy is severely limited and costly in many LMIC settings. These factors make it particularly unsuitable for diagnosing EED in children under five, where timely identification is critical (Watanabe & Petri, 2016). There has been no comprehensive review of studies utilising EED biomarkers in children under 5 years in LMIC settings. It is also not clear whether such studies rely on established biomarker cutoffs from HIC settings or have attempted to develop these in LMICs to assess pathogenic pathways in EED (Church et al., 2021).

1.3 Justification

There is need for a comprehensive review of the distribution of the biomarkers for EED in studies conducted in LMICs and determine whether any of these studies developed and reported optimal cutoffs for biomarkers in LMICs (Campbell et al., 2017; Espira et al., 2023; Syed et al., 2016). Secondly, for such studies, it's important to determine the utility and relevance of cutoffs for biomarkers where employed. Additionally, there is need to review such data and contextualise findings in relation to heterogeneity in assays, age groups, geographical differences, and study designs, among others for studies conducted in LMIC settings.

1.2 Objectives

1.2.1 Broad objective

To review the distribution of selected biomarkers of enteropathy: myeloperoxidase (MPO), calprotectin (CAL), neopterin (NEO), alpha-1-antitrypsin (AAT), and lactulose-to-mannitol ratio (L:M) in children under five years old living in LMICs.

1.2.1 Specific objectives

1. To review the distribution of biomarker concentrations among children under 5 years in LMICs.
2. To identify whether there are optimal cutoffs established in LMICs for use among children under 5 years.
3. To determine the proportion of children and or studies with elevated levels of the biomarkers based on cutoffs.

CHAPTER 2: LITERATURE REVIEW

2.0 Child Malnutrition in Low- and Middle-income countries

Undernutrition affects millions of children around the globe with recent estimates showing 148 million cases of stunting and 45 million instances of wasting (*Levels and Trends in Child Malnutrition*, 2023). According to the World Health Organization standards, stunting is defined as a height-for-age z-score of less than -2 standard deviations (HAZ < -2SD), while wasting is characterized by a weight-for-height z-score of less than -2 SD (WHZ < -2SD) (World Health Organization, 2006). Alarmingly, of the approximately 43% of stunted children, nearly 64 million, are found in Africa, alongside 12 million cases of childhood wasting. In comparison, Asia accounts for an even larger proportion, with around half of the world's stunted children and 70% of global wasting cases (*Levels and Trends in Child Malnutrition*, 2023).

Efforts to curb undernutrition and reduce mortality have guided the large-scale implementation of various nutritional and water, sanitation, and hygiene (WASH) interventions (Bhutta et al., 1999; Ceesay et al., 1997; Huybregts et al., 2012). A study done in 34 countries by Bhutta and colleagues to assess the effectiveness of these interventions in addressing undernutrition revealed that even if the existing nutrition-specific interventions were implemented globally with approximately 90% coverage, stunting prevalence would only be reduced by 20% (Bhutta et al., 2013). Kosek and colleagues highlighted that the expected underperformance is partly due to a complex interaction between infection and undernutrition (Bhutta et al., 2013; M. Kosek et al., 2014). Repeated exposure to enteric pathogens contributes to environmental enteric dysfunction (EED), characterized by intestinal inflammation, malabsorption, and increased intestinal permeability, which ultimately results in a vicious cycle where

undernutrition increases the risk of infection, while infection worsens undernutrition (Carvalho et al., 2025).

2.1 Mechanisms and complications in EED

EED is a complex, acquired, self-perpetuating, asymptomatic condition affecting the small bowel (Gerson et al., 1971). Its pathophysiology and aetiology are not well characterised and lack a definitive diagnostic criterion (Crane et al., 2015). It is believed to be the ramifications of an adaptive action to repeated environmental stimuli such as enteropathogens, which cause recurrent intestinal lesions, mucosal inflammation, and microbial translocation (Hodges et al., 2025). It is this repeated exposure that leads to a chronic state of immune activation at the intestinal epithelium, which changes the intestinal morphology and function in a number of ways (Marie et al., 2018; Salazar-Lindo et al., 2004).

The most observed morphological changes include crypt hyperplasia and villus atrophy, leading to an architecture resembling leaf-like or ridge structures, rather than the typical finger-like projections (Mathan, 1990). This results in reduced absorptive capacity of the gut. Increased epithelial permeability is a consequence of altered expression of tight junction proteins and brush border transporters (Abtahi et al., 2023). Notably, the barrier-forming tight-junction protein claudin-4 is significantly upregulated in populations characterized by EED (Abtahi et al., 2023). This upregulation, along with the increase in gut permeability, presents a paradox, as one would expect that such upregulation would serve to mitigate permeability in EED (Abtahi et al., 2023). Increased intestinal permeability, antigen translocation, chronic intestinal inflammation, malabsorption, hormonal disruption, and intestinal dysbiosis are highlighted as key interdependent mechanisms in EED that eventually result in poor health outcomes (Tickell et al., 2019).

Child stunting, micronutrient and macronutrient deficiencies, low vaccine efficacy, poor neurocognitive development, and increased child mortality have all been linked to EED (Gilmartin & Petri, 2015b; Loglo et al., 2025; Regassa et al., 2024; Semba et al., 2017). Linear growth faltering has been associated with fecal biomarkers of EED, such as MPO, NEO, and AAT (Arndt et al., 2016; R. Goto et al., 2009). With the persistently high global rates of child stunting, EED has become a key area in research aiming to understand the aetiology and reversal of stunting (Harper et al., 2018; Tickell et al., 2019).

2.2 Diagnosis of EED

2.2.1 Histology as the gold standard for the diagnosis of EED

The histological analysis of small intestine biopsies, the earliest approach to defining EED, is regarded as the gold standard in the diagnosis of EED (Fagundes-Neto et al., 1984; Keusch et al., 2013). The significant histologic findings include features of inflammatory enteropathy like decreased height and increased width of villi, hyperplasia of the crypts, elevated numbers of epithelial lymphocytes and mononuclear cells of the lamina propria(Mahfuz et al., 2024; Mathan, 1990). The plausible cause of increasing villus width is the expansion of the infiltrate of mononuclear inflammatory cells in the lamina propria (Hodges et al., 2021). However, the feasibility and ethical concerns of intestinal biopsies in children make it unsuitable in this population (Keusch et al., 2013). These reasons, together with the lack of a comprehensive, reproducible framework for characterising the small bowel lesions, explain the existing paucity of studies conducting histological examination of the small intestine (Liu et al., 2020).

Liu and colleagues introduced a histologic index aimed at identifying histological features that are specific to EED, along with those it shares with other enteropathies (Liu et al., 2020). This scoring index assigned a tiered categorical value, ranging from 4 to 5, to each

of the 11 histological parameters that represent various injury response mechanisms (Liu et al., 2020). The total score spans from 0 to 37. To account for instances where certain parameters could not be determined, a total histologic score percentage (TSP) was utilized (Liu et al., 2020). Kelly et al. (2024) modified this score by calculating TSP-5 from the original TSP by including only the scores for parameters found to be most discriminating for disease status and still producing a high area under the curve (AUC) (Kelly et al., 2024)

2.2.2 Biomarkers of intestinal permeability in EED

The dual sugar intestinal permeability test is a non-invasive technique used to measure both permeability and absorptive capacity of the small bowel mucosa (Cobden et al., 1978; Menzies et al., 1979). This test utilizes two different sugar molecules based on their molecular sizes and physiology of assimilation to assess the two aspects of the small intestines: the low molecular weight sugar reflecting the overall absorption capacity of the small intestines, while the larger one measures intestinal permeability (Weser & Sleisenger, 1965). Mannitol, or sometimes rhamnose, is used to evaluate absorption, whereas lactulose, or occasionally lactose, is used to assess permeability (Denno et al., 2016). The widely used combination is the lactulose to mannitol (L:M) test (Cobden et al., 1978).

The lactulose-to-mannitol (L:M) test is a non-invasive method used for decades to evaluate intestinal permeability and absorption in the small intestine. It measures the urinary excretion of two non-metabolizable sugar molecules, lactulose and mannitol, after oral administration. Mannitol (182Da) crosses the intestinal barrier via the transcellular route, while lactulose (342Da), being larger, uses the paracellular route. Factors like reduced villus height decrease mannitol absorption and lactulose permeability, whereas

increased intercellular space enhance lactulose permeation. The L:M ratio indicates these intestinal changes (Lunn et al., 1991). This test requires 2 or 5-hour urine collections following a standard dose of L:M mixture, which makes it time-consuming. Furthermore, the variability in procedures from one laboratory to another leads to inconsistencies (Musa et al., 2019). A study conducted in Malawi challenges the L:M interpretation by suggesting that using lactulose alone as a probe may provide more insight into gut health than L:M (Ordiz, Davitt, et al., 2018). However, it does not address the practical constraints associated with sugar absorption testing (Ordiz, Davitt, et al., 2018). It also lacks a clear interpretation of a low L:M ratio as both lactulose and mannitol have been shown to permeate the paracellular pathway, hence correlation of the two molecules (Lin A et al., 2020).

Alpha-1-antitrypsin (AAT) is a protein produced by liver cells that acts as a protease inhibitor in serum, shielding cells from inflammatory proteases (Maki et al., 1982). Its presence in stool indicates impaired intestinal permeability and potential protein-losing enteropathy, as it usually cannot cross the luminal barrier without significant dysfunction (Arndt et al., 2016). Since induced diarrhoea has been shown to cause an increase in AAT clearance, failure to consider it in establishing normal values may result in overdiagnosis of excess protein leakage in patients with diarrhoea (Strygler et al., 1990).

2.2.3 Biomarkers of intestinal inflammation in EED

Intestinal inflammation in EED has been consistently defined using faecal biomarkers such as myeloperoxidase (MPO), calprotectin (CAL), and neopterin (NEO) (Campbell et al., 2004; M. Kosek et al., 2013; Yui et al., 2003). Non-specific markers of intestinal inflammation, such as faecal leukocyte microscopy and faecal occult blood testing, were

frequently used before the advent of these biomarkers, but only in symptomatic diseases like diarrhoea and bacterial infections (Mercado et al., 2011; Seva-Pereira et al., 1994).

MPO (14kb) is one of the most salient enzymes of the subgroup peroxidase-cyclooxygenase (Arnhold, 2020). Resting neutrophils carry MPO in their azurophilic granules (Vanhamme et al., 2018). Upon activation of a neutrophil by uptake of a pathogen, an enzyme system of nicotinamide adenine dinucleotide phosphate (NADPH) oxidase enzymes is activated and recruited to the internal phagolysosomal membrane surface. At the same time, MPO is discharged into the phagosome. NADPH activity inside the phagosome forms superoxide anions (O_2^-) which are then transformed to hydrogen peroxide (H_2O_2) with the help of superoxide dismutase (SOD) enzyme. MPO subsequently activates the conversion of (H_2O_2) to reactive oxygen intermediates essential to innate immunity (Klebanoff et al., 2013). Its detection in stool serves as a specific marker for neutrophil activity, indicating inflammation within the intestinal mucosa (McCormick et al., 2017; Vanhamme et al., 2018).

NEO, an immune modulator, is a low molecular weight pteridine compound produced by macrophages or dendritic cells in response to stimulation by interferon-gamma (IFN-g) during pro-inflammatory responses mediated by Th1 lymphocytes (Huber et al., 1984). Elevated levels of NEO are seen in infection by viruses, intracellular living bacteria and parasites, autoimmune and other inflammatory conditions, malignant tumours, and allograft transplantation (Murr et al., 2002). Fahim and colleagues showed a strong association between NEO and Trichuris infection in children under 2 years, reflecting macrophage-driven inflammation (Fahim et al., 2018). It is used as an indicator of cellular immune activation in the gut.

Calprotectin (36kDa), also known as MRP8/14 (myeloid-related protein), calgranulin A/B, L1 protein, 27E10 antigen, and S100A8/A9, is a calcium- and zinc-binding protein of the S100 protein family first isolated from blood leukocytes (Vaos et al., 2013; Yui et al., 2003). Calprotectin accounts for 60% of the cytosolic proteins in neutrophils, and to a lesser extent monocytes and macrophages (Hinnouho GM et al., 2020; Smith et al., 1990). It plays a major role in inflammation and is a positive acute phase protein (Stříž & Trebichavský, 2004). It induces cell receptor expression involved in migration, adhesion, and phagocytosis of neutrophils, promotes chemotaxis and is implicated in the innate immune responses as a damage-associated molecular pattern (DAMP) protein (Stroncek et al., 2005). The presence of calprotectin in stool is a consequence of the migration of neutrophils into the intestinal tissue due to an infection or an inflammatory process (Yui et al., 2003).

2.3 Other approaches in the diagnosis of EED

Targeted transcriptomic analysis of host faecal messenger ribonucleic acid (mRNA) transcripts have also been used to predict the presence and severity of EED. In a Sierra Leone study, a quantitative, multiplexed, targeted transcriptomic analysis was conducted with droplet digital PCR (ddPCR) to measure the expression levels of 15 host faecal mRNAs and concluded that 8 of the mRNAs: regenerating family member 3 alpha (REG3A), gamma-interferon-inducible lysosomal thiol reductase (IFI30), aquaporin-9 (AQ9), 2,4-dienoyl-CoA reductase 1, (DECR1), selectin L (SELL), and baculoviral IAP repeat-containing 3 (BIRC3) could be used to identify EED based on percentage of lactulose (%L) with 74% specificity and 100% sensitivity (Singh et al., 2021). A previous study conducted in Malawi among 12- to 61-month-old children at risk of stunting also found 6 host faecal mRNA transcripts caudal type homeobox 1(CDX1), human leukocyte antigen class II DR alpha (HLA-DRA), mucin 12 (MUC12), regenerating family member

1 alpha (REG1A), tumour necrosis factor (TNF), and S100 calcium binding protein A8 (S100A8) to be able to identify EED measured using percentage of lactulose excreted in urine (%L), with 73% specificity and 84% sensitivity (Ordiz et al., 2016). Nonetheless, seven of these mRNAs failed to predict severe EED measured by %L in another study carried out in the same setting (Ordiz, Wold, et al., 2018).

The environmental enteropathy (EE) composite score, based on calculations of three biomarkers, AAT, MPO, and NEO, is increasingly being used to predict linear growth deficits (Becker-Dreps S et al., 2017; M. Kosek et al., 2013). This score, first proposed by Kosek and colleagues, was formulated on the percentile classes of these biomarkers (M. Kosek et al., 2013). This composite score, also referred to as the Environmental Enteropathy (EE) score, has been adapted by other researchers to include EndoCAb and AGP into the formula used to evaluate their relationship to stunting (Wirth et al., 2019) Campbell et al. (2017) challenged this practice, citing a relationship between MPO and NEO in this composite score and further highlighting the necessity to acquire more normative data and validation studies in paediatric samples (Campbell et al., 2017).

2.4 Validation of biomarkers of enteropathy

Studies have evaluated the validity of the L:M ratio as a measure of intestinal permeability relative to biopsies (R. Goto et al., 2009; Hossain et al., 2016; Iqbal et al., 2019; McCormick et al., 2017). A study performed on the Gambian infants revealed that there was no relationship between nutritional status and the L: M ratio (Lunn et al., 1991). However, it revealed a moderate positive correlation between mucosal B lymphocyte density and intraepithelial lymphocytes (IEL) (Lunn et al., 1991). It also showed a negative association between perforin-positive IEL and intraepithelial cells (IECs), indicating cytokine-mediated cell damage (Lunn et al., 1991). Another study involving

cohorts from Pakistan and Zambia also found no association between histological scores and the L:M ratio (Iqbal et al., 2019).

Recently, a multicountry study conducted in Zambia, Bangladesh, and Pakistan validated the L:M ratio, along with MPO, CAL, and NEO, against small bowel biopsies at 6 and 9 months of age (Mahfuz et al., 2024). The findings indicated that for each 1 unit increase in the IEL score, there was a corresponding 19% increase in calprotectin concentration (Mahfuz et al., 2024). Additionally, the presence of intramucosal Brunner's glands was linked to MPO and NEO, while chronic inflammation was associated with NEO (Mahfuz et al., 2024).

Paucity of validation studies and lack of standardization of these markers creates a state of incoherence in attempts to interpret findings on enteropathy diagnosis, evaluation of interventions targeting EED and malnutrition, or acute illnesses in children under 5 residing in low resource settings.

CHAPTER 3: METHODOLOGY

3.0 Study Design

This was a narrative review to examine research findings on biomarkers of EED in children under five years old living in LMICs. Additionally, the study included a synthesis of the existing literature as part of its design.

3.1 Literature search

A literature search was conducted on April 11, 2025, using PubMed, Google Scholar, and Web of Science. On PubMed, terms like "Environmental enteric dysfunction," "biomarker," and "children" were combined with "AND" and "OR." Google Scholar utilised the phrase "environmental enteric dysfunction" along with terms like "Myeloperoxidase," "AAT," "Calprotectin," or "Neopterin." Similar methods were applied to Web of Science with minor adjustments. No restrictions were applied on the date of publication or the type of article.

Table 1: Search Strategy

DATABASE	SEARCH STRING
PUBMED	((((((((((Environmental enteric dysfunction[Title/Abstract]) OR (Environmental enteropathy[Title/Abstract])) OR (Tropical enteropathy[Title/Abstract])) OR (Enteric dysfunction[Title/Abstract])) OR (Intestinal dysfunction[Title/Abstract])) OR (gut dysfunction[Title/Abstract])) OR (gut inflammation[Title/Abstract])) OR (enteropathy[Title/Abstract])) OR (malnutrition enteropathy[Title/Abstract])) OR (intestinal inflammation[Title/Abstract])) AND (((((((((((((((((((fecal biomarkers[Title/Abstract]) OR (stool biomarkers[Title/Abstract])) OR (EED biomarkers[Title/Abstract])) OR (Myeloperoxidase[Title/Abstract])) OR (fecal myeloperoxidase[Title/Abstract])) OR (MPO[Title/Abstract])) OR (fMPO[Title/Abstract])) OR (fecal MPO[Title/Abstract])) OR (Calprotectin[Title/Abstract])) OR (fecal calprotectin[Title/Abstract])) OR (fCP[Title/Abstract])) OR (fecal CP[Title/Abstract])) OR (Alpha-1-antitrypsin[Title/Abstract])) OR (Alpha 1-antitrypsin[Title/Abstract])) OR (AAT[Title/Abstract])) OR (fecal AAT[Title/Abstract])) OR (Neopterin[Title/Abstract])) OR (NEO[Title/Abstract])) OR (fNEO[Title/Abstract])) OR (fecal neopterin[Title/Abstract])) OR (fecal NEO[Title/Abstract]))) AND (((((children[Title/Abstract]) OR (child[Title/Abstract])) OR (infant[Title/Abstract])) OR (toddler[Title/Abstract])) OR (newborn[Title/Abstract]))
GOOGLE SCHOLAR	Biomarkers of environmental enteric dysfunction children biomarkers Myeloperoxidase OR AAT OR Calprotectin OR Neopterin "biomarkers of EED" -IBD -UC -CD
WEB OF SCIENCE	Biomarkers of environmental enteric dysfunction Myeloperoxidase OR AAT OR Calprotectin OR Neopterin OR L:M

3.2 Eligibility Criteria

3.2.1 Inclusion criteria

Studies focusing on environmental enteric dysfunction (EED) or malnutrition enteropathy in LMICs involving children under five years of age were included. Additionally, studies examining EED-related conditions in this population were considered if they assessed the association of biomarkers with factors such as stunting, micronutrient deficiencies, vaccine efficacy, and the effects of various interventions on these biomarkers.

3.2.2 Exclusion criteria

Studies involving participants over 5 years old, conducted in high-income countries, or assessing biomarkers under conditions other than EED or malnutrition enteropathy were excluded. Studies that did not report absolute biomarker concentrations or did not specify population averages were also excluded. Experimental studies on laboratory animals and those focusing on EED without measuring the specific biomarkers of interest were excluded. Publications written in languages other than English were also excluded.

3.3 Selection of studies

Covidence, an online tool for screening and selection in systematic reviews, was used. The screening process had three phases: first, Covidence's automated tools removed duplicates and ineligible studies. In the second phase, a two-reviewer panel screened titles and abstracts, excluding studies that did not focus on EED or EED-related conditions, particularly those addressing other conditions like Inflammatory Bowel Diseases. In addition, the reviewers manually checked and removed duplicates. For the third phase, articles were eligible for full-text screening if they met criteria on participant

characteristics, study setting, whether they assayed for the selected biomarkers, and reported the absolute concentrations. *Figure 1* shows a PRISMA diagram illustrating the screening and selection process.

3.4 Data Extraction

Eligible studies were exported from Covidence for offline data extraction using a structured Excel Microsoft software application. Information such as the author's name, year of publication, country, setting of the study (rural, urban, high-density, peri-urban, or mixed), study design, and participant characteristics (age group, number of participants, health status of children: community vs. hospitalised) was collected. Additionally, details on study arms or subgroups were recorded for interventional trials and case-control studies. Biomarker data were then extracted, including assay time-points, cutoff values, average concentrations (median and mean), and the lower (Q1) and upper (Q3) quartiles.

3.5 Data Visualisation

The collected data was visualised in R version 4.5.0 using the ggplot2, dplyr, and tidyr libraries. Forest plots illustrated the distribution of median concentrations, with Q1 as the lower limit and Q3 as the upper limit, while reference lines indicated the HIC cutoff from the literature. The Patchwork library combined the metadata with the forest plots. Visualisation of countries was done in Mapchart, an online tool for generating geographic maps

CHAPTER 4: RESULTS

4.0 Literature search.

A total of 402 publications were found through database searches, and 8 relevant articles were obtained via the "snowball" technique *(Figure 1*). After removing 40 duplicates and 176 ineligible references, 193 articles were screened. In-depth screening led to the exclusion of 148 studies that mainly focused on conditions like inflammatory bowel disease (IBD) and conditions other than environmental enteric dysfunction (EED). Forty-five articles were subjected to full-text screening, among which nine were excluded for the reasons highlighted in *Table 2.* Therefore, a total of 36 studies met the eligibility criteria and were included in this review (*Table 3*).

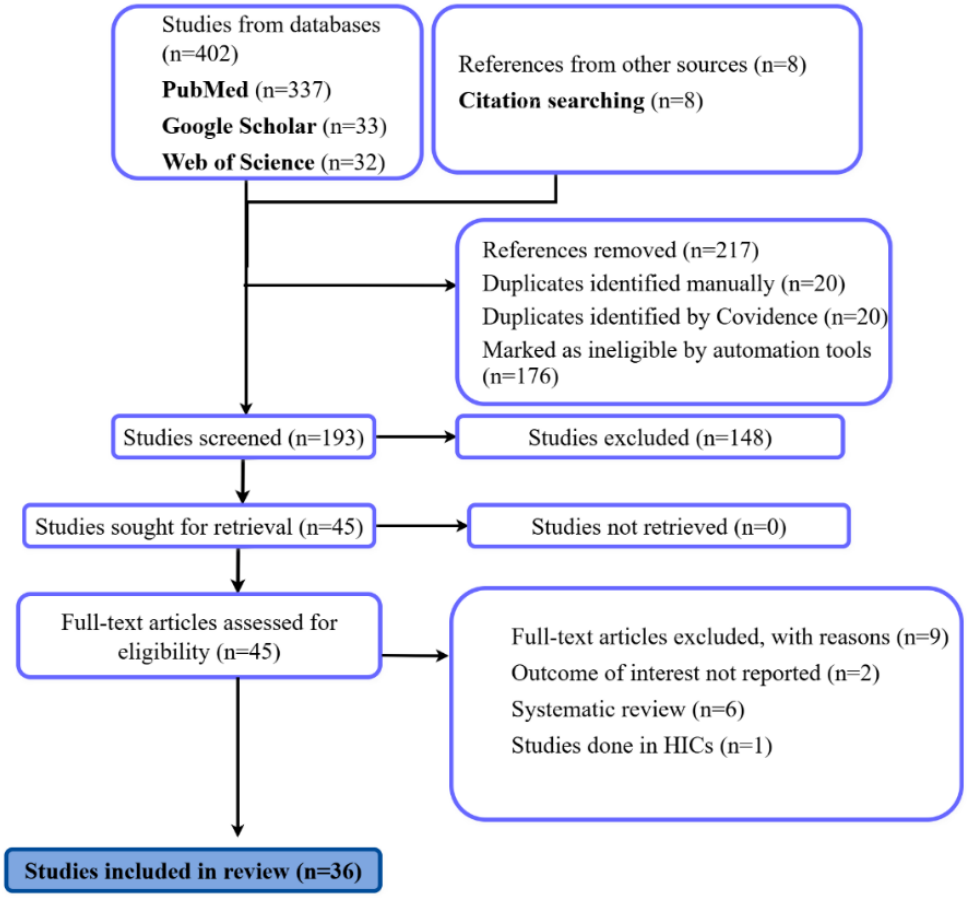

Figure 1: A PRISMA flow diagram illustrating the screening and selection of articles included in this review.

Table 2: Excluded studies that underwent full-text screening

SNo.	Author, Year	PMID	Reason for exclusion
1	Budge,2019	30753710	A narrative review
2	Mahfuz,2017	28801442	A study protocol
3	Pagani,2016	27529276	A study on Iron Deficiency Anaemia
4	Delawalla,2023		Systematic review
5	Tickel,2019	31760941	Systematic review
6	Mäki,1982	6279037	Study done in a high-income country
7	Etheredge,2018	29613921	Biomarkers of interest were not assayed.
8	Marie,2018	29988114	Narrative review
9	Rachmadi,2024	39230089	Systematic review

4.1 Characteristics of the included studies

Following a thorough evaluation of full-text articles, this review selected 36 studies that included 13 cohort studies, 13 randomised controlled trials, seven cross-sectional studies, and three case-control studies *(Table 3*). The studies were conducted in 22 countries, with Bangladesh and Malawi having the highest number of studies (*Figure 2*). The studies took place in various geographical settings: 22 in rural areas, 15 in urban areas, 4 in peri-urban areas, 5 in mixed residential settings, and 5 in high-density urban locations. Additionally, most of the studies were conducted among children in community settings, while 4 involved those in hospital settings (*Table 3*). The number of children included in the individual studies varied widely across sites, ranging from 17 to 1,512 participants. The children's ages ranged from 0 to 59 months, and the timing of the assays varied among the studies but included newborns (<1month), infants (2-9months), toddlers (< 2 years), and preschoolers (< 5 years). Of the 36 studies, 28 reported their average concentrations as medians with interquartile ranges (IQRs), while six reported means with or without standard deviations (SDs) and 95% confidence intervals (CIs).

Figure 2: Map of countries included in this review

The number of studies per country (shaded in dark gray) is as follows: South Asia; Bangladesh (13), Pakistan (3), India (3), Nepal (2), Indonesia (1), and Laos (1). Sub-Saharan Africa; Malawi (5), Burkina Faso (1), Madagascar (1), Kenya (1), Central African Republic (1), Sierra Leone (1), South Africa (1), Gambia (1), Zambia (1), Tanzania (2), Uganda (2), Zimbabwe (2), and Ethiopia (1). South and Central America; Brazil (1), Peru and (1), Nicaragua (1).

Table 3: List of studies included in this review and their metadata

Study	Aim	Cutoff	Study Design	ELISA kits used	Assay timepoints
1. *CHAIN*	Understand biological and social drivers of mortality among acutely ill children in sub-Saharan Africa and South Asia.	Not used	Cohort study	Not indicated	AAT, MPO, CAL assayed at discharge
2. *Lin et al., 2019*	Assess the effect of Water, Sanitation, and hygiene, and Nutrition on EED	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Cluster randomised controlled trial	Not indicated. L:M: High-performance liquid chromatography coupled with electrospray ionization tandem mass spectrometry	Ages of 3, 14, and 28 months respectively.
3. *Liu et al., 2022*	Test whether Lipid-based nutrient (LNS) supplements could reduce the levels of intestinal inflammation, repair, and permeability of children.	AAT <270µg/g REG1B	Retrospective Birth cohort study using data and samples from RCT for Lipid-based nutrient, or iron and folic acid, or multiple micronutrients during pregnancy and 6 months after delivery.	CAL (Hycult Biotech), AAT (PromoCell GmbH)	6, 18, and 30 months of age
4. *Arndt et al., 2016*	Describe the longitudinal patterns and contribution of EED to stunting in Bangladeshi children.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Retrospective data analysis from the Bangladesh site for the MAL-ED 2-year prospective observational study	MPO (Alpco, Salem, NH), NEO (GenWay Biotech, San Diego, CA), AAT (Biovendor, Candler, NC)	Ages 3, 6, 9, 12, 15, 18, and 21 months
5. *Fahim et al., 2018*	Association of intestinal pathogens with faecal markers of EED among slum-dwelling Bangladeshi children.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	The MAL-ED birth cohort data of Bangladesh site were used to conduct this analysis.	MPO (Alpco, Salem, New Hampshire), NEO (GenWay Biotech, San Diego, CA), AAT (Biovendor, Chandler, NC)	Ages 7, 15 and 24 months of age.
6. *Fahim et al., 2020*	Investigate factors associated with *H. pylori* infection, identify its relationship with faecal markers of EED, and indicators of the nutritional status among slum-dwelling Bangladeshi children.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L CAL<200µg/g	Retrospective analysis within the Bangladesh Environmental Enteric Dysfunction (BEED) study - a community-based nutrition intervention (daily egg, whole milk, micronutrients and counselling for 6 days in a week) among 12 to 18 months old children stunted or at risk of stunting for 90 days	AAT (Biovendor, Chandler, North Carolina), NEO (GenWay Biotech, San Diego, California), CAL (BÜHLMANN, Schönenbuch, Switzerland), MPO (Alpco, Salem, New Hampshire)	Non-diarrheal samples at baseline and after completion of the intervention.
7. *Gazi et al., 2020*	Association between infections with *E. coli* pathotypes, biomarkers of EED, and anthropometry among malnourished Bangladeshi children.	Not used	Data from the Bangladesh Environmental Enteric Dysfunction (BEED) study was used	MPO (Alpco, Salem, New Hampshire), NEO (GenWay, Biotech, San Diego, California), AAT (Biovendo Chandler, North Carolina), CAL (BUHLMANN, Schonenbuch, Switzerland),	Not indicated
8. *Hasan et al., 2020*	Explore the relationship between lipocalin-2 (LCN2) and low-density lipoprotein receptor-related protein-1 (LRP1) with gut inflammation and biomarkers of EED in malnourished Bangladeshi children.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Data from the Bangladesh Environmental Enteric Dysfunction (BEED) study was used	NEO (GenWay Biotech, California, USA), MPO (Alpco, New Hampshire, USA), AAT(Biovendor, North Carolina, USA)	Not indicated
9. *Kosek et al., 2013*	Test association between biomarkers of EED and linear growth among children in LMICs.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	MAL-ED Birth Cohort study	MPO (Alpco,Salem, NH), NEO (GenWay Biotech, San Diego, CA), AAT (Biovendor, Candler, NC)	15 days of age
10. *Long et al., 2019*	To compare zinc absorption in young Bangladeshi children screened for EED with L:M ratio. Compare biomarkers of intestinal and systemic inflammation and micronutrients status	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L CAL<50µg/g	A single-blind experimental design to measure zinc absorption from micronutrient powder.	CAL and MPO, (ALPCO Immunodiagnostics), NEO (Genway Biotech), AAT (Immunochrom GmbH)	Not indicated
11. *Mondal et al., 2019*	Measure absorption of zinc and examine its relationships with markers of inflammation and gut function.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L CAL<50µg/g	RCT on zinc absorption using dual stable isotope tracer and isotope dilution methods.	Not indicated	Not indicated
12. *Morseth et al., 2018*	Investigate associations between nutrient intake and EED among 9-24 months aged Nepalese children.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L L:M<0.07	The MAL-ED Nepal site provided data from children enrolled within 17 days from birth and followed at least until 24 months for the analyses	MPO (ALPCO, Salem, NH), AAT (BioVendor, Candler, NC), NEO (GenWay Biotech, San Diego, CA),	Urine samples at 15 months. Stools data at 12, 15, 18, 21 and 24 months.

13. Muller-Hauser et al., 2023	Effect of trained women's in gardening, poultry rearing, and food hygiene on EED in Bangladeshi children younger than 24 months.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Cluster-randomized trial enrolled 2,705 married women and their children younger than 3 years of age.	MPO (R&D Systems, Minneapolis, MN), NEO (IBL International, Germany), AAT (R&D Systems, Minneapolis, MN	Stool samples from children at trial end.
14. Gizaw et al., 2022	Assess faecal biomarkers of EED and associated factors among children aged 24–59 months in rural northwest Ethiopia	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Community-based cross-sectional study among 235 randomly selected children in a rural setting.	MPO (Immundiagnostik, AG, Germany), NEO (GenWay Biotech Inc., USA), AAT (BioVendor, ImmuChorm, Germany)	Between 24 to 59 months
15. Shivakumar et al., 2022	Estimate the Tryptophan rates for oxidation and conversion to Kynurenine in young slum dwelling Indian children.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L LRR ≥0.068	RCT among children aged 18–24 months with and without EED based on lactulose/rhamnose ratio (LRR).	MPO (EDI Epitope Diagnostics Inc.), NEO (Demeditec Diagnostics GmbH), AAT (ImmuChrom GmbH), CAL (EDI Epitope Diagnostics Inc.)	Stool
16. Sinharoy et al., 2021	Evaluate effect of a combined water and sanitation intervention in rural India on biomarkers of EED in children under five. Assess association of EED biomarkers with child anthropometry	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Sub-study nested within a water and sanitation intervention.	MPO (Alpco, Salem, NH), NEO (GenWay Biotech, San Diego, CA), AAT (µg/g) (Biovendor, Candler, NC)	Not indicated
17. Hinnouoho et al., 2020	Impact of supplemental zinc on EED biomarkers among young Laotian children	MPO<2000ng/mL NEO<70nmol/L CAL<50µg/g	RCT among children aged 6–23 months with zinc, or micronutrient, or therapeutic zinc for diarrhea and followed for 36 weeks.	NEO (GenWay Biotech, San Diego, CA), MP (ALPCO, Salem, NH), CAL (ALPCO, Salem, NH)	Stool samples at baseline and trial end
18. Crane et al., 2022	Examine intestinal dysbiosis and/or small intestinal bacterial overgrowth for the onset of EED and stunting during infancy.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Birth cohort of 100 infants followed for nine months where intestinal function and small intestinal bacterial overgrowth were assessed.	Not indicated	Monthly LM, breath hydrogen tests, and 16S sequencing- microbiota
19. Kortekangas et al., 2022	Assess the associations between gut microbiota and EED biomarkers in rural Malawian children.	AAT<270µg/g	Nutrition intervention trial	CAL (Hycult Biotech, Uden, The Netherlands), AAT (PromoCell, GmbH, Heidelberg, Germany),	Ages 6, 18, and 30 months
20. Versloot et al., 2018	Evaluate pathogen and diarrhea and duration of hospitalization among children with SAM from	CAL: 538µg/g (<6mo.), 214µg/g (6 mo.<3y.), 75 µg/g>3y.	Nested cohort among children from a randomized trial that compared outcomes of 3 commonly used WHO rehabilitation isocaloric diets with varying carbohydrate and fat ratios.	Not indicated	~21 months
21. Becker-Dreps et al., 2017	Examine the association between faecal biomarkers of EED and immunogenicity to RV5.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Cohort study	AAT(BioVendor, Asheville, NC), NEO (Genway Biotech Inc, San Diego, CA), MPO (ALPCO, Salem, NH), CAL (Eagle Biosciences, Nashua,NH)	2 months
22. Ahmed et al., 2022	Understand the association between RV IgA antibody seroconversion with child growth, enteropathogens, and biomarkers of EED in rural Pakistan.	MPO<2000ng/mL NEO<70nmol/L	Data from children enrolled in the Study of Environmental Enteropathy in Malnutrition (SEEM), a community-based interventional longitudinal trial, was used	Not indicated	at 6 months and 9 months
23. Singh et al., 2021	Develop host faecal mRNA transcripts score for EED and examine associations between the EED biomarkers and growth among children with MAM in Sierra Leone.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Nested cohort study in a cluster-randomized trial assessing four specialised nutritious foods (SNFs) to treat children 6 to 59 months with MAM.	Not indicated	
24. Chen et al., 2023	Investigate the associations between biomarkers of mycotoxin exposure, enteric infections, and enteropathy in Tanzania.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Nested within the Etiology, Risk Factors, and Interactions of Enteric Infections and Malnutrition and the Consequences for Child Health and Development (MAL-ED).	MPO (Alpco, Salem, NH), NEO (GenWay Biotech, San Diego, CA), AAT (Biovendor, Candler, NC)	24 months
25. Van Der Merwe et al., 2013	To test whether early omega-3 and long-chain polyunsaturated fatty acid supplementation improves infant intestinal integrity, growth, and cognitive function among rural Gambian infants.	CAL:538µg/g (1<6months) L:M<0.07	A randomized controlled trial conducted among children aged 3–9 mo.	CAL (Phical; CALPROAS)	3 months
26. Larenyo et al., 2020	Evaluate gut function of Ugandan children with SAM using faecal and blood biomarkers	MPO<2000ng/mL NEO<70nmol/L	A cross-sectional study, nested within a randomized trial at Mulago hospital, Uganda among subgroups of 400 children with complicated SAM and 30 community controls.	MPO ((R&D systems, Mineapolis, MN), NEO (IBL Salzburg, Austria)	Not indicated
27. Simuyandi et al., 2019	To investigate the relationships between specific enteric pathogens and stool markers of EED in Lusaka, Zambia.	Environmental Enteropathy (EE) score	Used clinical data and stored stool samples collected prior and during the early months of widespread rotavirus vaccine implementation.	CAL, MPO & AAT (Immundiagnostik AG, Bensheim Germany),	Not indicated
28. Vonaesch et al., 2022	Compare biomarkers of EED in children aged 2–5 years with and without stunting and test their associations with anaemia in Antananarivo, Madagascar and Bangui, Central African Republic	AAT <1.25mg/g dry or <0.15mg/g CAL<150mg/g (2-3y.),	A cross-sectional study on stunting among 460 children within the Afribiota project.	Not indicated	Not indicated

		<100mg/g (>3y) of faecal wet weight			
29. Fitria et al., 2025	Evaluate intestinal inflammation in malnourished children with symptomatic and asymptomatic norovirus infections in Surabaya, Indonesia	Not indicated	Cross-sectional study involving stool samples from children under five years old were analyzed.	NEO & CAL ELISA Kits from BTLab (Zhejiang, China)	Not indicated
30. Church et al., 2021	Determine association between EED and RVV immunogenicity and the effect of improved WASH on EED around the time of rotavirus vaccination.	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Nested cohort study	NEO (GenWay Biotech, San Diego, USA), MPO (Immundianostik, Germany), AAT (BioVendor, Brno, Czech Republic),	stool collected at 1, 3, 6, 12 and 18 months of age
31. Bartels et al., 2018	Tested therapeutic formula on improving enteropathy, tolerability and safety in Malawian children with complicated SAM	Not indicated	Randomised, open-label, 3-arm study conducted in children aged 6–23 months.	CAL (Bühlmann Schönenbuch, Switzerland), AAT (Immunodiagnostik, Hamburg, Germany),	Not specified (baseline and endline)
32. Campbell et al., 2018	Determine impact of EED and systemic inflammation on growth in rural Bangladeshi children.	Not indicated	Enrolled children in community-based randomised food supplementation trial.	L:M (HPLC (Dionex, Thermo Fisher, CA), MPO (ALPCO Diagnostics, Salem, NH), NEO (ALPCO Diagnostics, Salem, NH), AAT (BioVendor, LLC, Asheville, NC)	Samples at age 18 month
33. Gough EK et al., 2018	Investigate the impact of WASH and improved feeding on EED and describe the evolution of EED during infancy	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L L:M<0.07	Nested cohort	NEO (GenWay, San Diego,CA, USA), MPO (Immundianostik, Germany), AAT (BioVendor, Brno, Czech Republic)	Biomarkers evaluated at 1, 3, 6, 12, and 18 months of age.
34. Jannat et al., 2023	Investigate daily yogurt supplementation to reducing intestinal inflammation in 6-month-old Bangladeshi infants at risk of stunting over three months	AAT<270µg/g MPO<2000ng/mL NEO<70nmol/L	Three arms pilot randomized controlled trial	MPO (Immundiagnostic AG, Germany), NEO (GeneWay Biotech Inc, San Diego, CA AAT (Immuchrom, Biovendor, Chandler, NC)	Measured biomarkers before and after intervention
35. Colston et al., 2019	Identify clinically relevant biomarkers of EED that can inform intervention early in the disease process.	Not indicated	Peruvian birth cohort	Not indicated	samples at 7 and 15 months
36. Mutasa et al., 2021	Evaluate associations between EED biomarkers and linear growth velocity in rural Zimbabwe.	NEO, AAT, MPO, L:M	A subgroup of 1169 children enrolled in the Sanitation Hygiene Infant Nutrition Efficacy trial followed longitudinally from birth to 18 mo of age	MPO (Immundiagnostik, Germany), NEO (GenWay Biotech, Inc., San Diego, CA, USA), AAT (BioVendor, Brno, Czech Republic)	At infant ages 1, 3, 6, 12, and 18 months
Abbreviations:	EED- Environmental enteric dysfunction, LNS-Lipid-based Nutrient supplements, LCN2-Lipocalin-2, LRP1- Low density lipoprotein receptor related protein, RVV- Rotavirus Vaccine, IgA- Immunoglobulin A, SAM- severe acute malnutrition, MAM- Moderate acute malnutrition, mRNA-messenger ribonucleic acid, RCT-randomised controlled trial, MAL-ED- Etiology, Risk Factors, and Interactions of Enteric Infections and Malnutrition and the Consequences for Child Health and Development, SNF- Specialised nutritious foods, SEEM- the Study of Environmental Enteropathy in Malnutrition, BEED- Bangladesh Environmental Enteric Dysfunction, LRR- Lactulose/Rhamnose Ratio, MPO- myeloperoxidase, NE0- neopterin, CAL- calprotectin, AAT- alpha-1-antitrypsin, ELISA- enzyme-linked immunosorbent assay, L:M- lactulose/mannitol ratio, R&D- research and diagnostics, USA- United States of America, CA- Carlifornia, NH- New Hampshire, NC- North Carolina, GmbH- *Gesellschaft mit beschränkter Haftung*.				

4.3 Distribution of Biomarkers of intestinal permeability and absorption in LMICs

4.3.1 Faecal Alpha-1-Antitrypsin (AAT)

Twenty-six studies in 19 LMIC countries evaluated AAT among 12,253 children, as shown in *Figure 3.* The studies were conducted in 18 rural, 15 urban, 5 high-density urban, 4 peri-urban, and 3 mixed residence areas. Even though concentrations were comparable across settings, children in the community had higher frequencies of elevated AAT. None of the studies established cutoffs for AAT within LMIC settings across the different age groups. Twenty-one studies used a single cutoff of <270µg/g for intestinal permeability derived from HICs, with 84% of the 19 studies reporting median concentrations exceeding this threshold (Thomas et al., 1981). Only 2 of the seven studies that reported means were below this cutoff.

Unlike newborns and infants, toddlers had higher frequencies of elevated median levels, implying higher intestinal permeability. Longitudinal AAT levels at 6-, 18-, and 30-month showed reductions with age in Malawian children (Kortekangas E et al., 2022). However, a study assessing AAT at 3, 14, and 28 months found that AAT levels rose at 14 months and fell at 28 months. (Lin A et al., 2020). The lowest concentration noted was 20 µg/g in a rural Malawi cohort (Kortekangas E et al., 2022). In contrast, a study in Zimbabwe reported levels 18 times higher than the cutoff, at 4860 µg/g, with an upper quartile of 13,950 µg/g (Simuyandi et al., 2019).

4.3.2 Lactulose to Mannitol Ratio (L:M)

Nine studies in 6 LMIC countries evaluated L:M ratio among 8,329 children (*Figure 4*). The studies were conducted in 6 rural and 3 peri-urban settings. L:M permeability testing was only done in community children. Three studies used a cutoff of 0.07, first defined in LMIC setting among Guatemalan infants (K. Goto et al., 1999). When using this cutoff,

this study found a 30% prevalence of increased intestinal permeability and reduced absorptive capacity in these infants. This review found 83% of the studies reporting median concentrations above this threshold. Only 1 of the three studies that reported mean concentrations was below this cutoff. There was no defined pattern of distribution across settings or age groups.

Infants and toddlers have the highest frequency of L:M ratios above the cutoff of 0.07. A Gambian trial assessing the effect of long-chain polyunsaturated fatty acids supplementation on gut integrity, growth, and cognitive function among infants aged 3-9 months recorded values twice the cutoff in both control and intervention groups (van der Merwe LF et al., 2013). A study in Zimbabwe by Gough et al. (2020) reported mean concentrations of 0.84 and 0.49 at 6 and 18 months, respectively, which were above this cutoff.

StudyID	Country	StudtDesign	N	Age	Setting	Hospitalised	Timepoint	Median	Q1	Q3
Simuyandi,2019_27	Zambia	Cohort	234	Toddlers	Mixed	Hospitalized	baseline	4860.00	1210.00	13950.00
Sing,2021_23	Sierra Leone	Cohort	200	Pre-school*	Rural	Community	at enrollment	2217.49	1756.43	2947.04
Kosek,2013_9e	Peru	Cohort	131	Newborns	Rural	Community	<1month	600.00	310.00	990.00
Kosek,2013_9c	India	Cohort	192	Newborns	Urban	Community	<1month	590.00	290.00	1120.00
Kosek,2013_9d	Nepal	Cohort	241	Newborns	Urban	Community	<1month	590.00	310.00	1120.00
Shivakumar,2022_15b	India	Cohort	27	Toddlers	Urban	Community	Not indicated	590.00	380.00	800.00
Shivakumar,2022_15a	India	Cohort	19	Toddlers	Urban	Community	Not indicated	530.00	190.00	970.00
Long,2019_10a	Bangladesh	RCT	33	Toddlers	Peri-urban	Community	baseline	520.00	230.00	1030.00
Hasan MM,2021_8	Bangladesh	cross-sectional	140	Toddlers	Urban (dense)	Community	Not indicated	490.00	230.00	760.00
Kosek,2013_9a	Bangladesh	Cohort	211	Newborns	Urban	Community	<1month	470.00	250.00	820.00
Lin, 2019_2b	Bangladesh	cRCT	1499	Toddlers#	Rural	Community	14 mo	450.00	250.00	890.00
Long,2019_10b	Bangladesh	RCT	33	Toddlers	Peri-urban	Community	baseline	440.00	210.00	750.00
Muller-Hauser,2023_13	Bangladesh	cRCT	574	Toddler*	Rural	Community	post-intervention	422.00	233.00	710.00
Morseth,2018_12b	Nepal	Cohort	144	Newborns	Peri-urban	Community	Not indicated	420.00	230.00	860.00
Mondal,2019_11a	Bangladesh	RCT	20	Toddlers	Peri-urban	Community	baseline	400.00	150.00	790.00
Gazi,2022_7e	Bangladesh	RCT	1009	Toddlers	Urban	Community	Not indicated	400.00	160.00	720.00
Gazi,2022_7g	Bangladesh	RCT	840	Toddlers	Urban	Community	Not indicated	400.00	160.00	730.00
Gazi,2022_7j	Bangladesh	RCT	582	Toddlers	Urban	Community	Not indicated	400.00	160.00	730.00
Morseth,2018_12a	Nepal	Cohort	211	Newborns	Peri-urban	Community	Not indicated	400.00	210.00	700.00
Lin, 2019_2a	Bangladesh	cRCT	1090	Infants#	Rural	Community	3 mo	390.00	230.00	790.00
Gazi,2022_7b	Bangladesh	RCT	717	Toddlers	Urban	Community	Not indicated	390.00	160.00	720.00
Gazi,2022_7c	Bangladesh	RCT	583	Toddlers	Urban	Community	Not indicated	390.00	160.00	700.00
Gazi,2022_7d	Bangladesh	RCT	459	Toddlers	Urban	Community	Not indicated	390.00	160.00	740.00
Lin, 2019_2c	Bangladesh	cRCT	1512	Preschool#	Rural	Community	28 mo	380.00	220.00	720.00
Arndt, 2016_4	Bangladesh	Cohort	246	Newborns	Urban (dense)	Community	<1month	380.00	190.00	717.50
Gazi,2022_7a	Bangladesh	RCT	325	Toddlers	Urban	Community	Not indicated	380.00	160.00	710.00
Gazi,2022_7h	Bangladesh	RCT	202	Toddlers	Urban	Community	Not indicated	380.00	140.00	680.00
Gazi,2022_7i	Bangladesh	RCT	460	Toddlers	Urban	Community	Not indicated	380.00	150.00	670.00
Mutasa,2021_36b	Zimbabwe	Cohort	523	Infants#	Rural	Community	3 mo	360.00	180.00	780.00
Gizaw,2022_14	Ethiopia	Cross-sectional	235	Preschool*	Rural	Community	Not indicated	350.00	268.00	385.00
Morseth,2018_12c	Nepal	Cohort	205	Newborns	Peri-urban	Community	Not indicated	350.00	230.00	600.00
Mutasa,2021_36a	Zimbabwe	Cohort	472	Newborns	Rural	Community	1 mo	350.00	170.00	980.00
Mutasa,2021_36c	Zimbabwe	Cohort	820	Infants#	Rural	Community	6 mo	340.00	190.00	640.00
Crane,2022_37	Kenya	C/sec	99	Newborns	Rural	Community	<1month	332.48	132.85	696.59
Fahim, 2018_5	Bangladesh	Cohort	222	Toddler*	Urban (dense)	Community	7,15,and 24 mo	330.00	180.00	620.00
Sinharoy,2021_16	India	Case-control	471	Pre-school*	Rural	Community	2 timepoints	330.00	164.10	595.40
Gazi,2022_7f	Bangladesh	RCT	33	Toddlers	Urban	Community	Not indicated	330.00	140.00	510.00
Kosek,2013_9h	Tanzania	Cohort	57	Newborns	Rural	Community	<1month	310.00	180.00	610.00
Morseth,2018_12d	Nepal	Cohort	207	Newborns	Peri-urban	Community	Not indicated	310.00	160.00	610.00
Mutasa,2021_36d	Zimbabwe	Cohort	898	Toddlers#	Rural	Community	12 mo	300.00	160.00	580.00
Kosek,2013_9b	Brazil	Cohort	42	Newborns	Urban	Community	<1month	290.00	140.00	620.00
Morseth,2018_12e	Nepal	Cohort	192	Newborns	Peri-urban	Community	Not indicated	270.00	140.00	530.00
CHAIN_1e	Malawi	Cohort	282	Pre-school*	Urban	Hospitalized	at discharge	250.29	123.99	504.73
Kosek,2013_9g	South Africa	Cohort	38	Newborns	Mixed	Community	<1month	250.00	190.00	630.00
Kosek,2013_9f	Pakistan	Cohort	257	Newborns	Rural	Community	<1month	230.00	120.00	520.00
Mutasa,2021_36e	Zimbabwe	Cohort	940	Toddlers#	Rural	Community	18 mo	230.00	110.00	420.00
CHAIN_1f	Pakistan	Cohort	314	Pre-school*	Urban	Hospitalized	at discharge	227.07	128.23	446.79
CHAIN_1i	Burkina Faso	Cohort	311	Pre-school*	Urban	Hospitalized	at discharge	217.55	105.88	458.93
CHAIN_1b	Kenya	Cohort	244	Pre-school*	Urban (dense)	Hospitalized	at discharge	185.97	80.06	364.82
CHAIN_1a	Kenya	Cohort	257	Toddlers#	Rural	Hospitalized	at discharge	180.82	96.14	387.93
CHAIN_1d	Uganda	Cohort	429	Pre-school*	Urban	Hospitalized	at discharge	179.35	82.75	338.02
CHAIN_1g	Bangladesh	Cohort	493	Pre-school*	Urban	Hospitalized	at discharge	178.43	66.84	349.73
Mondal,2019_11b	Bangladesh	RCT	20	Toddlers	Peri-urban	Community	baseline	170.00	120.00	350.00
CHAIN_1c	Kenya	Cohort	399	Pre-school*	Rural	Hospitalized	at discharge	161.25	88.97	283.36
Becker-Dreps,2017_21b	Nicaragua	Cohort	19	Infants	Urban	Community	Preimmunization	152.30	104.10	224.30
CHAIN_1h	Bangladesh	Cohort	333	Pre-school*	Rural	Hospitalized	at discharge	145.00	69.60	336.75
Becker-Dreps,2017_21a	Nicaragua	Cohort	24	Infants	Urban	Community	Preimmunization	129.50	87.30	164.80
Kortekangas,2022_19a	Malawi	Nested cohort	459	Infants#	Rural	Community	6 mo	100.00	60.00	210.00
Kortekangas,2022_19b	Malawi	Nested cohort	578	Toddlers#	Rural	Community	18 mo	50.00	30.00	80.00
Liu, 2022_3	Malawi	Cohort	620	Toddlers#	Rural	Community	18 mo	49.00	26.00	790.00
Kortekangas,2022_19c	Malawi	Nested cohort	532	Toddler#	Rural	Community	30 mo	20.00	10.00	50.00

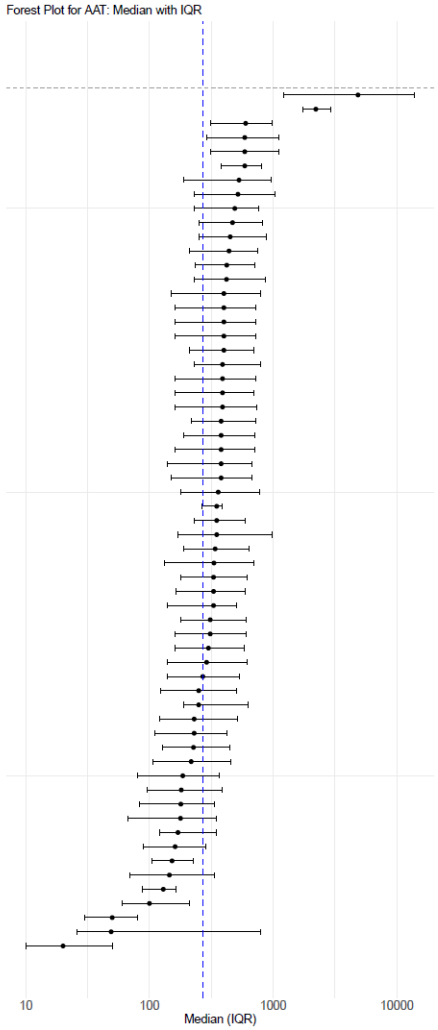

Figure 3: Forest Plot illustrating the distribution of AAT in LMIC

The StudyID was derived as follows: *author name*, *year of publication* _ *Study number* in **Table 3**. Studies with more than one study arms e.g RCTs, or different assay timepoints, were assigned an additional letter to the study number (e.g. 34a, 34b). Q1- 25th percentile, Q3- 75th percentile. The blue dotted line illustrates the AAT cutoff of 270µg/g. *Age groups with overlapping ages were defined by the upper age group. #Age groups defined by assay time-point.

StudyID	Country	StudtDesign	N	Age	Setting	Hospitalised	Timepoint	Median	Q1	Q3
Mondal,2019_11a	Bangladesh	RCT	18 to 20	Toddlers	Peri-urban	Community	baseline	0.16	0.08	0.23
Van Der Merwe,2013_25a	The Gambia	RCT	85	Infants	Rural	Community	NA	0.15	0.10	0.22
Van Der Merwe,2013_25b	The Gambia	RCT	82	Infants	Rural	Community	NA	0.15	0.12	0.20
Mutasa,2021_36a	Zimbabwe	Cohort	430	Infants#	Rural	Community	3 mo	0.14	0.06	0.53
Mutasa,2021_36c	Zimbabwe	Cohort	546	Infants#	Rural	Community	12 mo	0.13	0.06	0.42
Mutasa,2021_36b	Zimbabwe	Cohort	533	Infants#	Rural	Community	6 mo	0.11	0.06	0.33
Sing,2021_23	Sierra Leone	Cohort	422	Preschool*	Rural	Community	at enrollment	0.09	0.06	0.15
Mondal,2019_11b	Bangladesh	RCT	17 to 20	Toddlers	Peri-urban	Community	baseline	0.09	0.04	0.15
Morseth,2018_12	Nepal	Cohort	144	Toddlers#	Peri-urban	Community	15 mo	0.07	0.05	0.12
Mutasa,2021_36d	Zimbabwe	Cohort	627	Toddlers#	Rural	Community	18 mo	0.07	0.04	0.16
Lin, 2019_2b	Bangladesh	cRCT	1499	Toddlers#	Rural	Community	14 mo	0.05	0.03	0.08
Lin, 2019_2a	Bangladesh	cRCT	1090	Infants#	Rural	Community	3 mo	0.04	0.03	0.06
Lin, 2019_2c	Bangladesh	cRCT	1512	Toddlers#	Rural	Community	28 mo	0.03	0.02	0.05

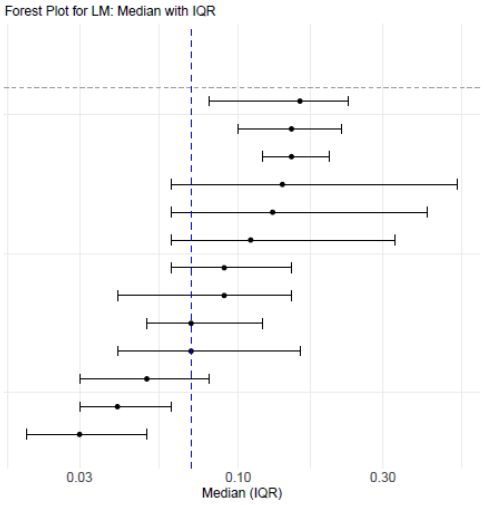

Figure 4: Forest Plot illustrating the distribution of L:M in LMICs

The StudyID was derived as follows: *author name, year of publication _Study number* in **Table 3**. Studies with more than one study arms e.g RCTs, or different assay timepoints, were assigned an additional alphabet suffix to the study number (e.g. 34a, 34b) for plotting. RCT- Randomised controlled trials, cRCTs- Cluster randomised controlled trials, c/sec- cross-sectional, N- sample size, Timepoint- the period when the biomarker was assayed, Q1- 25th percentile, Q3- 75th percentile. The blue dotted line illustrates the L:M ratio cutoff of 0.07.

4.2 Distribution of biomarkers of intestinal inflammation in LMICs

4.2.1 Myeloperoxidase (MPO)

Twenty-seven studies in 17 LMIC countries assessed MPO among 14,463 children, as shown in *Figure 5*. The studies were conducted in 19 rural, 13 urban, five urban high-density settings, three peri-urban, and three mixed residential areas. MPO concentrations appeared to be relatively similar across all settings. Hospitalised children had lower concentrations compared to their community counterparts, although Lanyero et al. (2019) in their study among severely acute malnourished (SAM) children recorded high MPO concentrations (Lanyero B et al., 2019).

No study has attempted to generate MPO cutoffs in the LMIC setting. Twenty-two (22) studies utilised a cutoff of <2000 ng/mL which was derived from Japanese IBD cases as an indicator of intestinal inflammation, with 81% of the 21 studies reporting median concentrations exceeding this threshold (Saiki, 1998). Only one of the seven studies reported a mean below the HIC cutoff. Median concentrations were mainly skewed towards elevation, with newborns and infants having the highest elevated levels, while toddlers and preschoolers tended towards concentrations within the cutoff. Only Nicaraguan infants had median concentrations below the cutoff (Becker-Dreps S et al., 2017). In one study that measured MPO at 3, 14, and 28 months, an age-defined decrease was noticed (Lin A et al., 2020). The lowest median MPO was 668 ng/mL in Bangladesh, while a study in India reported levels 35 times higher than the HIC cutoff (Mondal et al., 2019; Shivakumar N et al., 2022).

StudyID	Country	StudtDesign	N	Age	Setting	Hospitalised	Timepoint	Median	Q1	Q3
Shivakumar,2022_15b	India	Case-ctrl	27	Toddlers	Urban	Community	Not indicated	71668.00	50815.00	95192
Shivakumar,2022_15a	India	Case-ctrl	19	Toddlers	Urban	Community	Not indicated	59221.00	43233.00	80226
Sing,2021_23	Sierra Leone	Cohort	200	Preschool	Rural	Community	at enrollment	42172.51	19253.97	87815
Muller-Hauser,2023_13	Bangladesh	cRCT	574	Toddlers*	Rural	Community	post-intervention	23233.00	9481.00	5522
Lanyero,2019_26a	Uganda	Nested C/sec	349	Preschool*	Mixed	Hospitalised	1 timepoint	18083.00	5904.00	67128
Kosek,2013_9b	Tanzania	Cohort	65	Newborns	Rural	Community	<1 mo	17949.77	9612.64	26330
Kosek,2013_9g	South Africa	Cohort	74	Newborns	Mixed	Community	<1 mo	16284.92	6530.56	25171
Kosek,2013_9c	India	Cohort	96	Newborns	Urban	Community	<1 mo	14574.97	6093.03	27507
Kosek,2013_9d	Nepal	Cohort	234	Newborns	Urban	Community	<1 mo	14484.40	7499.47	25317
Kosek,2013_9e	Peru	Cohort	59	Newborns	Rural	Community	<1 mo	11623.52	5765.75	21883
Lin, 2019_2a	Bangladesh	cRCT	1090	Infants	Rural	Community	3 mo	9593.81	4125.47	19683
Kosek,2013_9a	Bangladesh	Cohort	194	Newborns	Urban	Community	<1 mo	8838.23	4552.95	15020
Ahmed,2022_22b	Pakistan	RCT	115	Infants#	Rural	Community	6 mo	8800.00	3050.00	18000
Ahmed,2022_22a	Pakistan	RCT	149	Infants#	Rural	Community	6 mo	8725.50	2900.00	19535
Kosek,2013_9f	Pakistan	Cohort	156	Newborns	Rural	Community	<1 mo	8452.01	3950.82	12868
Mutasa,2021_36b	Zimbabwe	Cohort	531	Newborns	Rural	Community	1 mo	8221.00	3962.00	18074
Lanyero,2019_26b	Uganda	Nested C/sec	25	Preschool*	Urban	Community	1 timepoint	7482.00	3837.00	12373
Mutasa,2021_36c	Zimbabwe	Cohort	821	Infants#	Rural	Community	3 mo	7233.00	4166.00	12413
Crane,2022_37	Kenya	C/sec	99	Newborns	Rural	Community	< 1mo	6950.00	2539.00	20591
Kosek,2013_9b	Brazil	Cohort	38	Newborns	Urban	Community	<1 mo	6847.91	3884.41	12452
Morseth,2018_12a	Nepal	Cohort	211	Infants	Peri-Urban	Community	12 mo	5935.20	2524.50	14267
Mutasa,2021_36a	Zimbabwe	Cohort	478	Toddlers#	Rural	Community	24 mo	5861.00	2777.00	11647
Lin, 2019_2b	Bangladesh	cRCT	1499	Toddlers#	Rural	Community	14 mo	5540.75	2728.83	10311
Ahmed,2022_22d	Pakistan	RCT	115	Infants#	Rural	Community	9 mo	4623.30	2189.00	11650
Mutasa,2021_36d	Zimbabwe	Cohort	896	Infants#	Rural	Community	6 mo	3959.00	2107.00	7044
Fahim, 2018_5	Bangladesh	Cohort	222	Toddlers	Urban (dense)	Community	7, 15, and 24 mo	3895.42	1563.79	8432
Morseth,2018_12b	Nepal	Cohort	144	Toddlers#	Peri-Urban	Community	15 mo	3883.80	1880.00	6660
Arndt, 2016_4	Bangladesh	Cohort	246	Newborns	Urban (dense)	Community	<1 mo	3354.90	1594.90	7430
CHAIN_1e	Malawi	Cohort	282	Preschool*	Urban	Hospitalised	at discharge	3351.75	1343.75	8500
Gizaw,2022_14	Ethiopia	C/sec	235	Toddlers	Rural	Community	1 timepoint	3320.20	1890.00	6528
Morseth,2018_12c	Nepal	Cohort	205	Toddlers#	Peri-Urban	Community	18 mo	3162.50	1184.50	5693
Becker-Dreps,2017_21b	Nicaragua	Cohort	19	Infants	Urban	Community	Preimmunization	3100.00	1800.00	400
Ahmed, 2022_22c	Pakistan	RCT	149	Infants#	Rural	Community	9 mo	3050.00	1250.00	7587
Hinnouoho, 2020_17	Laos	RCT	720	Toddlers	Rural	Community	baseline	2710.00	1249.00	7305
CHAIN_1f	Pakistan	Cohort	314	Preschool*	Urban	Hospitalised	at discharge	2690.25	1061.50	6414
Simuyandi,2019_27	Zambia	Case-ctrl	234	Toddlers	Mixed	Community	baseline	2560.40	806.10	6522
CHAIN_1a	Kenya	Cohort	257	Preschool*	Rural	Hospitalised	at discharge	2553.25	1123.88	6397
Gazi,2022_7h	Bangladesh	RCT	202	Toddlers	Urban	Community	Not indicated	2490.00	1170.00	6400
Lin, 2019_2c	Bangladesh	cRCT	1512	Toddlers#	Rural	Community	28 mo	2429.83	1273.31	4743
CHAIN_1i	Burkina Paso	Cohort	311	Preschool*	Urban	Hospitalised	at discharge	2420.50	1139.75	6550
CHAIN_1g	Bangladesh	Cohort	493	Preschool*	Urban	Hospitalised	at discharge	2411.75	961.38	5953
Gazi,2022_7j	Bangladesh	RCT	582	Toddlers	Urban	Community	Not indicated	2410.00	1260.00	5050
Gazi,2022_7b	Bangladesh	RCT	717	Toddlers	Urban	Community	Not indicated	2340.00	1170.00	5460
Gazi,2022_7c	Bangladesh	RCT	583	Toddlers	Urban	Community	Not indicated	2200.00	1030.00	5510
Gazi,2022_7e	Bangladesh	RCT	1009	Toddlers	Urban	Community	Not indicated	2200.00	1150.00	5130
CHAIN_1c	Kenya	Cohort	399	Preschool*	Rural	Hospitalised	at discharge	2143.50	885.00	5019
CHAIN_1d	Uganda	Cohort	429	Preschool*	Urban	Hospitalised	at discharge	2121.00	1007.50	4852
Gazi,2022_7d	Bangladesh	RCT	459	Toddlers	Urban	Community	Not indicated	2120.00	1200.00	4640
Gazi,2022_7g	Bangladesh	RCT	840	Toddlers	Urban	Community	Not indicated	2100.00	1130.00	4600
Mutasa,2021_36e	Zimbabwe	Cohort	941	Infants#	Rural	Community	12 mo	2063.00	967.00	4109
Morseth,2018_12d	Nepal	Cohort	207	Toddlers#	Peri-Urban	Community	21 mo	2001.70	854.30	3976
Gazi,2022_7f	Bangladesh	RCT	33	Toddlers	Urban	Community	Not indicated	1950.00	1140.00	3600
Gazi,2022_7i	Bangladesh	RCT	460	Toddlers	Urban	Community	Not indicated	1950.00	968.00	5100
Gazi,2022_7a	Bangladesh	RCT	325	Toddlers	Urban	Community	Not indicated	1940.00	1030.00	4360
CHAIN_1b	Kenya	Cohort	244	Preschool*	Urban (dense)	Hospitalised	at discharge	1909.50	875.00	5410
Long,2019_10b	Bangladesh	RCT	27 to 37	Toddlers	Peri-Urban	Community	baseline	1750.00	763.00	3515
Hasan MM,2021_8	Bangladesh	Nested C/sec	222	Toddlers	Urban (dense)	Community	1 timepoint	1720.00	796.00	3720
Morseth,2018_12e	Nepal	Cohort	192	Toddlers#	Peri-Urban	Community	24 mo	1541.60	592.90	2983
CHAIN_1h	Bangladesh	Cohort	333	Preschool*	Rural	Hospitalised	at discharge	1394.50	630.00	3185
Long,2019_10a	Bangladesh	RCT	33	Toddlers	Peri-Urban	Community	baseline	1317.00	862.00	2740
Mondal,2019_11a	Bangladesh	RCT	18 to 20	Toddlers	Peri-Urban	Community	baseline	1161.00	421.00	3383
Becker-Dreps,2017_21a	Nicaragua	Cohort	24	Infants	Urban	Community	Preimmunization	1100.00	500.00	190
Sinharoy,2021_16	India	Nested Case-ctrl	471	Preschool	Rural	Community	2 timepoints	874.88	528.00	1731
Mondal,2019_11b	Bangladesh	RCT	17 to 20	Toddlers	Peri-Urban	Community	baseline	668.00	144.00	2635

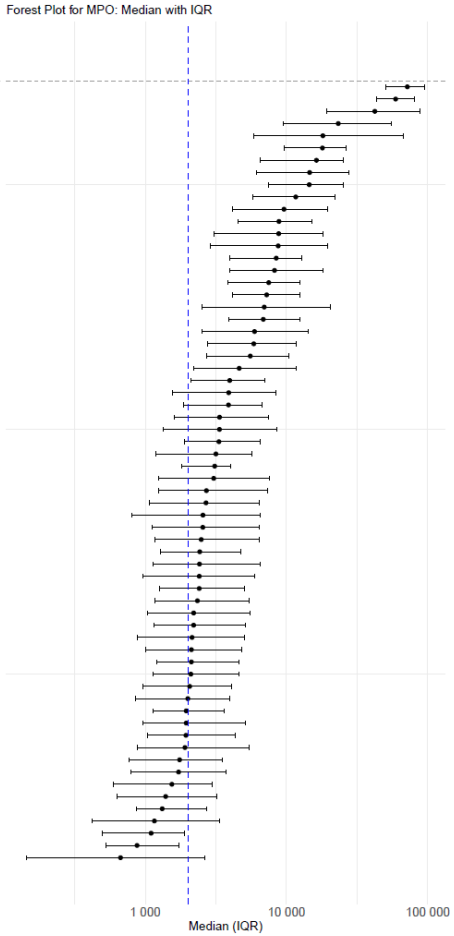

Figure 5: Forest Plot illustrating the distribution of MPO in LMICs

The StudyID was derived as follows: *author name, year of publication _Study number* in **Table 3**. Studies with more than one study arms e.g RCTs, or different assay timepoints, were assigned an additional alphabet to the study number (e.g. 34a, 34b) for plotting. The blue dotted line illustrates the MPO cutoff of 2000ng/mL. *Age groups with overlapping ages were defined by the upper age group. #Age groups defined by assay time-point.

4.2.2 Faecal Calprotectin (CAL)

Sixteen studies in 10 LMIC countries assessed CAL levels among 6926 children as shown in *Figure 6*. The studies were conducted in 6 rural, 10 urban, 2 urban (high-density), 2 peri-urban, and 3 mixed residential areas. Generally, urban settings appeared to have median concentrations above the cutoff compared to peri-urban and rural settings. Concentrations were similarly high in hospitalised and community children.

One study generated age-specific CAL cutoffs: 249mg/kg in 0< 1-year-olds, 75mg/kg in 1< 4-year-olds, and 28mg/kg in 4< 12-year-old Ugandan children (Hestvik et al., 2011). There was no harmonised cutoff across studies, as some used thresholds indicated in ELISA kits used (often 50µg/g, and rarely 200 µg/g) while others used age-specific cutoffs derived from HIC studies. For example, Vonaesch et al. (2022) used 150 and 100 mg/g of faecal wet weight for children aged 2 to 3 years, and above 3 years, respectively, which was considered normal in a University Hospital in Paris, France (Vonaesch et al., 2022). The European standard of less than 263mg/kg for 3-month-old children was used in Gambian infants (van der Merwe et al., 2013). This study found that 85% of the infant concentrations were above this cutoff (van der Merwe et al., 2013). Versloot et al. (2018) used European age-specific cutoffs of 538mg/kg (1-6months), 214mg/kg (6months to 3years), and 75mg/kg (3<4years) (Oord & Hornung, 2014). Fahim and colleagues also found 85% and 77% of baseline and post-intervention participant concentrations, respectively, to be above the European standard of <200µg/g used in their study (Fahim SM et al., 2020). Similarly, this review found 71% of the 14 studies reported median concentrations exceeding a 150µg/g threshold, mainly used in HICs to indicate active IBD (K. Kolho et al., 2012). Two of the four studies that reported mean instead of median values were above these cutoff (Bartels RH et al., 2019; Liu Z et al., 2022). Among the

six studies focusing on children over one year, 67% reported levels above 150 µg/g. The lowest median concentration was 52 µg/g in a newborn cohort from rural Malawi, while the highest was 1029 µg/g in a Gambian trial (Kortekangas E et al., 2022; van der Merwe LF et al., 2013). Generally, urban settings appeared to have median concentrations above the cutoff compared to peri-urban and rural settings.

4.2.3 Faecal Neopterin (NEO)

Twenty-three studies in 17 LMIC countries evaluated NEO concentrations among 11,120 children, as in *Figure 7.* The studies were conducted in 8 urban, 16 rural, 4 urban (high-density), 3 peri-urban, and 2 mixed residence areas. Overall, concentration levels were high across all settings for hospitalised and community children. Lanyero and colleagues reported concentrations that were lower than those of most community children (Lanyero B et al., 2019). Even though there was no cutoff for NEO derived in LMICs, twenty-one studies used a cutoff of 70nmol/L derived from the HICs to indicate Th1 cell activity in the intestinal mucosa (Ledjeff et al., 2001). All 15 studies reported median concentrations above this cutoff. In contrast, only 1 out of the 8 studies that reported mean concentrations fell below this threshold.

While NEO was generally high in all children, most findings were in toddlers. The lowest median concentration recorded was 210 nmol/L in a case-control study conducted in urban Uganda involving children aged 6 to 59 months (Lanyero B et al., 2019). Conversely, pronounced concentration of 3997.2 nmol/L was seen in the MAL-ED study in South African newborns (M. Kosek et al., 2013).

StudyID	Country	StudtDesign	N	Age	Setting	Hospitalised	Timepoint	Median	Q1	Q3
Van Der Merwe,2013_25b	Gambia	RCT	83	Infants	Rural	Community	NA	1058.00	610.00	1639.00
Van Der Merwe,2013_25a	Gambia	RCT	82	Infants	Rural	Community	NA	1029.00	627.00	1534.00
Fahim,2020_6	Bangladesh	Cohort	319	Toddlers	Urban (dense)	Community	baseline	598.30	300.10	1041.50
Gazi,2022_7h	Bangladesh	RCT	202	Toddlers	Urban	Community	NA	584.00	338.00	886.00
Gazi,2022_7d	Bangladesh	RCT	459	Toddlers	Urban	Community	NA	555.00	289.00	920.00
Gazi,2022_7j	Bangladesh	RCT	582	Toddlers	Urban	Community	NA	529.00	307.00	880.00
Gazi,2022_7b	Bangladesh	RCT	717	Toddlers	Urban	Community	NA	522.00	289.00	911.00
Vonaesch,2022_28a	Madagascar	C/sec	202	Preschool*	Urban	Community	1 timepoint	515.00	54.00	10600.00
Gazi,2022_7e	Bangladesh	RCT	1009	Toddlers	Urban	Community	NA	501.00	275.00	871.00
Gazi,2022_7g	Bangladesh	RCT	840	Toddlers	Urban	Community	NA	485.00	260.00	847.00
Gazi,2022_7a	Bangladesh	RCT	325	Toddlers	Urban	Community	NA	477.00	230.00	795.00
Gazi,2022_7i	Bangladesh	RCT	460	Toddlers	Urban	Community	NA	470.00	230.00	840.00
Gazi,2022_7c	Bangladesh	RCT	583	Toddlers	Urban	Community	NA	467.00	249.00	819.00
Gazi,2022_7f	Bangladesh	RCT	33	Toddlers	Urban	Community	NA	452.00	226.00	839.00
Vonaesch,2022_28b	Madagascar	C/sec	215	Preschool*	Urban	Community	1 timepoint	404.00	55.00	4530.00
Kortekangas,2022_19a	Malwai	Nested cohort	459	Infants#	Rural	Community	6 mo	395.00	189.00	643.00
Versloot,2018_20a	Malawi	RCT	26	Preschool*	Mixed	Hospitalised	after stabilization	383.00	149.00	903.00
CHAIN_1e	Malawi	Cohort	282	Preschool*	Urban	Hospitalised	at discharge	334.65	150.82	1110.90
Vonaesch,2022_28a	CAR	C/sec	171	Preschool*	Urban	Community	1 timepoint	309.00	72.00	4600.00
Vonaesch,2022_28b	CAR	C/sec	216	Preschool*	Urban	Community	1 timepoint	303.00	56.00	4050.00
CHAIN_1f	Pakistan	Cohort	314	Preschool*	Urban	Hospitalised	at discharge	272.10	134.87	749.85
CHAIN_1i	Burkina Faso	Cohort	311	Preschool*	Urban	Hospitalised	at discharge	271.76	129.41	706.62
CHAIN_1g	Bangladesh	Cohort	493	Preschool*	Urban	Hospitalised	at discharge	270.10	95.08	709.76
CHAIN_1a	Kenya	Cohort	257	Preschool*	Rural	Hospitalised	at discharge	263.76	105.23	668.25
Shivakumar,2022_15b	India	Case-ctrl	27	Toddlers	Urban	Community	NA	244.00	112.00	740.00
CHAIN_1c	Kenya	Cohort	399	Preschool*	Rural	Hospitalised	at discharge	241.84	99.47	555.62
CHAIN_1b	Kenya	Cohort	244	Preschool*	Urban (dense)	Hospitalised	at discharge	224.55	107.61	607.62
CHAIN_1d	Uganda	Cohort	429	Preschool*	Urban	Hospitalised	at discharge	213.08	112.22	563.52
Becker-Dreps,2017_21b	Nicaragua	Cohort	19	Infants	Urban	Community	Preimmunization	199.10	158.50	321.80
CHAIN_1h	Bangladesh	Cohort	333	Preschool*	Rural	Hospitalised	at discharge	158.08	81.50	477.84
Becker-Dreps,2017_21a	Nicaragua	Cohort	24	Infants	Urban	Community	Preimmunization	156.20	152.00	174.60
Long,2019_10b	Bangladesh	RCT	33	Toddlers	Peri-Urban	Community	baseline	153.00	42.00	488.00
Versloot,2018_20b	Malawi	RCT	21	Preschool*	Mixed	Hospitalised	after stabilization	140.00	71.00	300.00
Shivakumar,2022_15a	India	Case-ctrl	19	Toddlers	Urban	Community	NA	135.00	91.00	490.00
Hinnouho, 2020_17	Lao	RCT	720	Toddlers*	Rural	Community	baseline	133.00	47.00	279.00
Long,2019_10a	Bangladesh	RCT	33	Toddlers	Peri-Urban	Community	baseline	128.00	93.00	261.00
Mondal,2019_11a	Bangladesh	RCT	18	Toddlers	Peri-Urban	Community	baseline	115.00	34.00	337.00
Kortekangas,2022_19b	Malwai	RCT	578	Toddlers#	Rural	Community	18 mo	107.00	5.00	293.00
Simuyandi,2019_27	Zambia	Case-ctrl	234	Toddlers	Mixed	Hospitalised	NA	79.10	0.00	362.50
Mondal,2019_11b	Bangladesh	RCT	17	Toddlers	Peri-Urban	Community	baseline	64.00	33.00	110.00
Kortekangas,2022_19c	Malwai	RCT	532	Preschool#	Rural	Community	30 mo	52.00	24.00	139.00

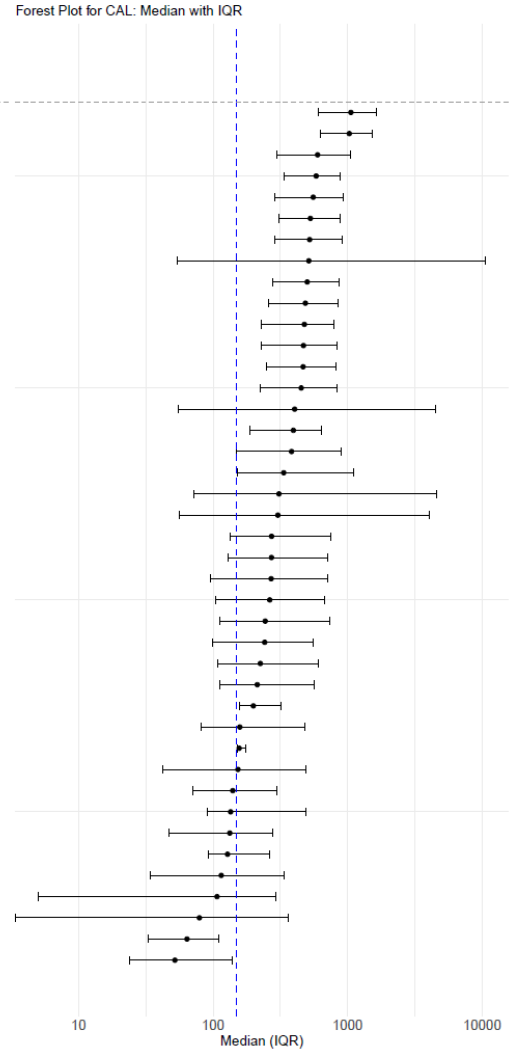

Figure 6: Forest Plot illustrating the distribution of CAL in LMICs

The StudyID was derived as follows: *author name*, *year of publication* _*Study number* in **Table 3**. Studies with more than one study arms e.g RCTs, or different assay timepoints, were assigned an additional alphabet to the study number (e.g. 34a, 34b). The blue dotted line illustrates the CAL cutoff of 150µg/g. *Age groups with overlapping ages were defined by the upper age group. #Age groups defined by assay time-point.

StudyID	Country	StudtDesign	N	Age	Setting	Hospitalised	Timepoint	Median	Q1	Q3
Kosek,2013_9g	South Africa	Cohort	74	Newborns	Mixed	Community	< 1mo	3997.17	2492.34	5830.18
Morseth,2018_12a	Nepal	Cohort	211	Infants#	Peri-urban	Community	12 mo	2439.70	1476.30	3511.20
Kosek,2013_9b	Brazil	Cohort	38	Newborns	Urban	Community	< 1mo	2385.39	1895.55	3303.96
Ahmed,2022_22a	Pakistan	RCT	149	Infants	Rural	Community	6 mo	2100.00	1037.50	3502.60
Kosek,2013_9f	Pakistan	Cohort	156	Newborns	Rural	Community	< 1mo	2076.65	1305.33	3820.75
Kosek,2013_9c	India	Cohort	96	Newborns	Urban	Community	< 1mo	2009.31	1417.08	2969.13
Ahmed,2022_22c	Pakistan	RCT	149	Infants	Rural	Community	9 mo	1975.00	927.40	2625.00
Kosek,2013_9e	Peru	Cohort	59	Newborns	Rural	Community	< 1mo	1884.87	1396.77	2827.90
Gazi,2022_7c	Bangladesh	RCT	583	Toddlers	Urban	Community	Not indicated	1860.00	858.00	3190.00
Ahmed,2022_22b	Pakistan	RCT	115	Infants	Rural	Community	6 mo	1834.50	990.00	2650.00
Ahmed,2022_22d	Pakistan	RCT	115	Infants	Rural	Community	9 mo	1800.00	875.00	2850.00
Gazi,2022_7g	Bangladesh	RCT	840	Toddlers	Urban	Community	Not indicated	1750.00	843.00	3070.00
Kosek,2013_9h	Tanzania	Cohort	196	Newborns	Rural	Community	< 1mo	1748.40	1155.42	2960.52
Gazi,2022_7i	Bangladesh	RCT	460	Toddlers	Urban	Community	Not indicated	1660.00	788.00	2990.00
Gazi,2022_7a	Bangladesh	RCT	325	Toddlers	Urban	Community	Not indicated	1650.00	824.00	2860.00
Gazi,2022_7e	Bangladesh	RCT	1009	Toddlers	Urban	Community	Not indicated	1650.00	826.00	2970.00
Gazi,2022_7b	Bangladesh	RCT	717	Toddlers	Urban	Community	Not indicated	1640.00	811.00	2990.00
Gazi,2022_7j	Bangladesh	RCT	582	Toddlers	Urban	Community	Not indicated	1640.00	843.00	2930.00
Morseth,2018_12b	Nepal	Cohort	144	Toddler#	Peri-urban	Community	15 mo	1549.50	822.50	2438.80
Muller-Hauser,2023_13	Bangladesh	Cluster RCT	574	Toddlers*	Rural	Community	post-intervention	1534.00	753.00	3089.00
Gazi,2022_7d	Bangladesh	RCT	459	Toddlers	Urban	Community	Not indicated	1490.00	755.00	2680.00
Mutasa,2021_36b	Zimbabwe	Cohort	516	Infants#	Rural	Community	3 mo	1442.00	867.00	2187.00
Kosek,2013_9a	Bangladesh	Cohort	194	Newborns	Urban	Community	< 1mo	1422.35	741.38	2237.37
Mutasa,2021_36c	Zimbabwe	Cohort	812	Infants#	Rural	Community	6 mo	1419.00	961.00	2153.00
Kosek,2013_9d	Nepal	Cohort	234	Newborns	Urban	Community	< 1mo	1413.73	1008.67	2179.72
Sinharoy,2021_16	India	Nested case-ctrl	471	Preschool*	Rural	Community	2 timepoints	1379.00	483.30	2626.80
Morseth,2018_12c	Nepal	Cohort	205	Toddler#	Peri-urban	Community	18 mo	1378.10	678.30	2318.50
Gazi,2022_7h	Bangladesh	RCT	202	Toddlers	Urban	Community	Not indicated	1250.00	719.00	2610.00
Hasan MM,2021_8	Bangladesh	C/sec	222	Toddlers	Urban (dense)	Community	1 timepoint	1240.00	567.00	2690.00
Gazi,2022_7f	Bangladesh	RCT	33	Toddlers	Urban	Community	Not indicated	1240.00	535.00	2850.00
Shivakumar,2022_15a	India	Case-ctrl	19	Toddlers	Urban	Community	Not indicated	1151.00	791.00	2849.00
Shivakumar,2022_15b	India	Case-ctrl	27	Toddlers	Urban	Community	Not indicated	1107.00	728.00	2362.00
Arndt, 2016_4	Bangladesh	Cohort	246	newborns	Urban (dense)	Community	<1 mo	1017.60	366.20	2210.80
Sing,2021_23	Sierra Leone	Cohort	200	Preschool*	Rural	Community	at enrollment	977.91	469.45	1878.22
Mutasa,2021_36d	Zimbabwe	Cohort	890	Infants#	Rural	Community	12 mo	977.00	461.00	1730.00
Morseth,2018_12d	Nepal	Cohort	207	Toddler#	Peri-urban	Community	21 mo	957.10	496.40	1874.90
Mutasa,2021_36a	Zimbabwe	Cohort	475	Newborns	Rural	Community	1 mo	927.00	560.00	1413.00
Fahim, 2018_5	Bangladesh	Cohort	222	Toddlers*	Urban (dense)	Community	7, 15, and 24 mo	890.81	331.57	2089.04
Morseth,2018_12e	Nepal	Cohort	192	Toddler#	Peri-urban	Community	24 mo	683.10	375.90	1266.30
Hinnouho, 2020_17	Lao	RCT	720	Toddlers*	Rural	Community	baseline	629.00	176.00	1829.00
Lanyero,2019_26a	Uganda	Nested case-ctrl	365	Preschool*	Mixed	Hospitalised	1 timepoint	541.00	274.00	1168.00
Crane,2022_37	Kenya	C/sec	99	Newborns	Rural	Community	<1 mo	426.00	162.50	1136.00
Becker-Dreps,2017_21b	Nicaragua	Cohort	19	Infants	Urban	Community	Preimmunization	412.40	334.00	747.50
Becker-Dreps,2017_21a	Nicaragua	Cohort	24	Infants	Urban	Community	Preimmunization	408.70	235.90	946.00
Mutasa,2021_36e	Zimbabwe	Cohort	940	Toddlers#	Rural	Community	18 mo	318.00	123.00	763.00
Lanyero,2019_26b	Uganda	Nested case-ctrl	25	Preschool*	Urban	Community	1 timepoint	210.00	103.00	372.00

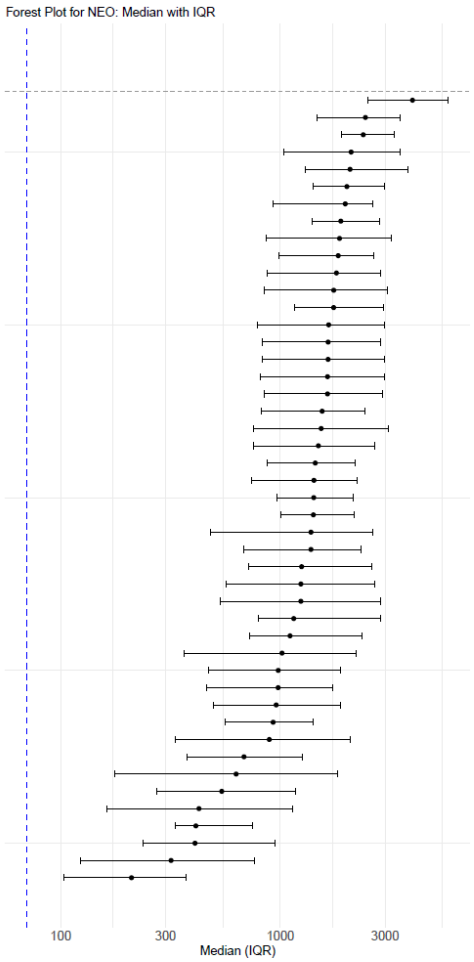

Figure 7: Forest Plot illustrating the distribution of NEO in LMICs

The StudyID was derived as follows: *author name*, *year of publication* _*Study number* in **Table 3**. Studies with more than one study arms e.g RCTs, or different assay timepoints, were assigned an additional alphabet to the study number (e.g. 34a, 34b). The blue dotted line illustrates the NEO cutoff of 70nmol/L. *Age groups with overlapping ages were defined by the upper age group. #Age groups defined by assay time-point.

CHAPTER 5: DISCUSSION

This review consolidates data regarding the distribution of biomarkers used to indicate environmental enteric dysfunction (EED) in low- and middle-income countries (LMICs). The study focused on enteric permeability as determined by AAT and L:M ratio and inflammation as measured by MPO, CAL and NEO, which are part of the key features of EED. The study identified 36 original studies that assessed these biomarkers among 17,430 children under 5 years old in LMICs. These studies were conducted in different geographical settings among children of varying age groups. This study found that, except for the L:M ratio and CAL, cutoffs used for other biomarkers are derived from HIC settings (K. Goto et al., 1999; Hestvik et al., 2011). Using these cutoffs, this review shows that children in LMIC settings often have elevated biomarker levels for permeability and intestinal inflammation. Cohort studies, such as those done in Bangladesh and Malawi, revealed age-dependent fluctuations in faecal marker concentrations.

This review highlights the widespread usage of biomarkers for diagnosing enteric permeability and inflammation in EED in LMIC settings, especially in the last decade. However, it is surprising that for most biomarkers, few studies have attempted to develop and establish cutoffs relevant for such biomarkers in these settings. Cutoffs used for AAT, MPO, and NEO were adopted from HIC settings. For example, the faecal AAT cutoff used was derived from 39 children from the USA (Thomas et al., 1981), while that for MPO was derived from 32 Japanese adults with inflammatory bowel disease (IBD) and 15 standard controls (Saiki, 1998). The NEO cutoff was derived from an Austrian study evaluating neopterin concentrations in colon dialysate (Ledjeff et al., 2001). In that study, 30 healthy adult controls on their regular diet (mean age 40.3±13.3yrs) were compared with 41 healthy volunteers on a starvation diet (mean age 56.0±9.5yrs), and eight patients

(3 acute nervous gastritis, one mild herpes labialis, and four common cold) (Ledjeff et al., 2001). The 70nmol/L mean concentration in the 30 adult controls has since been used as the normal upper cut-point for faecal NEO evaluation in EED studies across the LMICs (M. Kosek et al., 2013). The CAL cutoff of <200µg/g used by Fahim and colleagues likely originates from a European meta-analysis of 8 studies on calprotectin optimal cutoffs in paediatric IBD (Henderson et al., 2014). One of the studies by Henderson et al. (2012) revealed that while a cutoff of 50µg/g provided an excellent sensitivity, it had poor specificity (Henderson et al., 2012). They also observed a steady increase in specificity with increasing CAL levels and reaching a plateau at around 200-300µg/g (Henderson et al., 2012, 2014). CAL cutoffs used by Versloot and colleagues were generated in a European study involving 75 normal children as controls and 157 children with IBD in Denmark (Oord & Hornung, 2014).

This review notes that cutoffs for CAL, a marker of neutrophil activity in the intestinal mucosa, and the lactulose-to-mannitol (L:M) ratio, a marker of intestinal permeability and absorption, were generated in the LMIC setting (K. Goto et al., 1999; Hestvik et al., 2011). A study among 302 healthy Ugandan children aged 0 to 12 found consistent CAL levels with findings from HICs (Hestvik et al., 2011). Hestvik and colleagues generated age-specific cutoffs of 249mg/kg in 0< 1-year-olds, 75mg/kg in 1< 4-year-olds, and 28mg/kg in 4< 12-year-olds. However, they did not look at the feeding practices of children under 1 year, which might influence CAL levels in this age group (Dorosko et al., 2008). In 158 Guatemalan infants, the 5-hour urine L:M ratio was measured and compared by age group, feeding practices, anthropometric index, and serum iron and zinc concentrations (K. Goto et al., 1999). A cutoff of <0.068 was obtained (K. Goto et al., 1999). The wide usage of the biomarkers necessitates studies to derive cutoffs relevant to

LMIC settings for MPO, AAT, and NEO and to validate the few studies where such cutoffs have been derived.

AAT and L:M ratio have been used to assess intestinal permeability and absorption. Elevations were seen in 84% and 83% of the studies assaying AAT and L:M, respectively. Concentrations were generally high across settings, likely confirming that EED is ubiquitous in the LMICs (Keusch et al., 2013). Community children exhibited higher concentrations of AAT, likely due to the high enteropathogen burden in community settings (M. N. Kosek et al., 2017). A study done in Zambia and Zimbabwe among children with severe acute malnutrition (SAM) found AAT to normalise with hospitalisation, indicating improved gut barrier function (Sturgeon et al., 2023). However, this was not observed in another study conducted in Kenya and Pakistan among hospitalised children (Tickell et al., 2023).

Toddlers had higher faecal AAT levels as illustrated by its distribution within each age category (Mondal et al., 2019; Shivakumar N et al., 2022; Simuyandi et al., 2019; Singh et al., 2021). These findings show an impaired gut barrier and a reduced absorptive capacity of the gut mucosa across settings in the LMIC (Lunn et al., 1991). Previous studies have shown that AAT levels rise in the second year of life, partly explained by increased contact of the child with the environment-harboured pathogens due to heightened mobility (McCormick et al., 2017). Pathologic causes, like enteropathogen exposures that may vary with geographic locations which could additionally influence the distributions observed (Watanabe & Petri, 2016). Fahim et al. (2020) observed an inverse association of H.pylori with AAT, indicating gut damage in *Helicobacter pylori* infection (Fahim SM et al., 2020).

High proportions of studies with elevated MPO, NEO, and CAL highlight widespread intestinal inflammation in children under five across all LMIC settings (Becker-Dreps S et al., 2017; Chen C et al., 2023; Church et al., 2021; Fitria AL et al., 2025; Hinnouho GM et al., 2020; Lanyero B et al., 2019). Intestinal inflammation markers were generally high in children in the community. An increase in MPO without NEO indicates intestinal inflammation, while an increase in NEO alone may indicate normal gut immunity (Campbell et al., 2017). Ahmed et al. (2022) found that non-seroconverted Pakistani infants had pronounced MPO and NEO concentrations both at 6 months and 9 months, showing that this inflammatory response begins early in infancy (Ahmed et al., 2022). Elevated MPO and NEO among hospitalised SAM children in Uganda were likely due to intestinal pathogens contributing to intestinal inflammation (Becker-Dreps S et al., 2017). Cessation of exclusive breastfeeding was associated with increased intestinal inflammation, as shown by the high MPO and NEO among Kenyan infants (Crane et al., 2022). Breastfeeding has been associated with reduced permeability and increased gut integrity (Tickell et al., 2023). In Ethiopia, elevations were associated with open defecation, mouthing of soil-contaminated materials, *Escherichia coli* contamination of drinking water, food, and soil, and intestinal parasites (Gizaw et al., 2022). In India, high concentrations were associated with slow kynurenine synthesis (Shivakumar N et al., 2022).

CAL concentrations in the LMICs were high across all settings, indicating increased intestinal inflammation (Yui et al., 2003). Even though elevations were seen in all settings, urban areas had high counts of elevated concentrations. This finding appears to contradict the hypothesis that EED is more prevalent in rural settings due to low hygiene standards (M. Kosek et al., 2013; Praharaj et al., 2018). However, urban areas that are densely populated with low hygiene standards are likely to have higher intestinal

inflammation markers due to repeated enteropathogen exposures in such settings. This observation is likely biased by the high number of studies reporting CAL which conducted in urban settings. Distribution pattern across ages revealed that CAL reduces with age, similar to findings in other studies (Hestvik et al., 2011; Oord & Hornung, 2014). Apart from age, dietary intake, breastfeeding status, and enteropathogen exposure may determine CAL concentrations (McCormick et al., 2017). In breastfeeding children, higher breast milk intake may contribute to higher concentrations of calprotectin (Koivusaari et al., 2023; Park et al., 2020). Fecal calprotectin levels in children older than one year are generally low and comparable between children and adolescents (K.-L. Kolho & Alfthan, 2020). Van Der Merwe and colleagues found 85% of the Gambian infants to be above the European <200µg/g cutoff (van der Merwe LF et al., 2013). They observed that supplementation with long-chain polyunsaturated fatty acids did not improve intestinal inflammation in EED (van der Merwe LF et al., 2013). Enteric infections like enterotoxigenic *Escherichia coli* usually elevate CAL levels in children in the LMICs (Gazi et al., 2022).

While these biomarkers can be easily evaluated using commercially available kits, a meta-analysis establishing CAL reference range in children aged 0 to 4 found that variability may arise from differences in kits used (Zeng et al., 2025). They found that ELISA kits by different manufacturers yielded wide variability in CAL levels (Zeng et al., 2025). Faecal sampling methods also influence biomarker concentrations. Symptomatic diarrheal stool samples, or faecal sampling after diarrheal episodes, affect concentrations of these biomarkers (Morseth, 2019). Stool sampling after ingesting known laxative compounds such as lactulose and mannitol may alter biomarker concentrations (Ordiz, Davitt, et al., 2018).

This is the first review to assess the distribution of biomarkers of EED in the LMICs against the available cutoffs utilised in these settings. It adds to the existing knowledge on EED distribution and the currently used biomarker cutoffs in EED diagnosis. However, the proportions provided here are not enough to translate into EED prevalence in the LMICs.

CHAPTER 6: CONCLUSION

The findings in this review highlight the generally high levels of EED biomarkers among children under five in LMICs based on the HIC standards. Interventions targeting EED could have population-wide impacts on growth and development in LMICs. They also emphasise the need to establish cutoffs relevant for children in these settings. Furthermore, standardised procedures should be developed to reduce variability in findings and enable harmonisation and meta-synthesis of findings across studies.

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